Praktikum Machine Learning - Multiple Features

Multiple Feature, Overfitting, dan Perbaikannya

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Tujuan Setelah praktikum ini, mahasiswa diharapkan mampu:

1. memahami penggunaan **multiple feature**
2. membedakan **underfitting** dan **overfitting**
3. memperbaiki overfitting dengan **feature selection**
4. melihat dampak **penambahan data**

Petunjuk

- Jalankan cell dari atas ke bawah.
- Kode utama sudah disediakan, fokus praktikum pada **observasi hasil** dan **pemahaman konsep** bukan pada coding.
- Submit file `.ipynb` dan `.pdf` melalui SPADA.
- Penamaan file: NIM_NAMA_multi_fitur.[ipynb/pdf]

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

RANDSEED = 42
np.random.seed(RANDSEED)
plt.rcParams["figure.figsize"] = (7,4)
plt.rcParams["axes.grid"] = True

#Predefined Code
def add_bias(x):
  bias = np.ones(x.shape[0])
  return np.c_[bias, x]

def cost(y,pred, w=None,lamda=0.0):
    m = y.shape[0]
    mse = ((pred - y) ** 2).sum() / (2 * m)
    reg = 0
    if w is not None and lamda!=0.0:
      reg = (lamda / (2 * m)) * (w[1:] ** 2).sum()   # w0 tidak ikut
    return mse + reg

def predict(w,x):
  return x @ w

def update_bobot(w,xb,y,alpha, lamda=0.0):
```

```
  output = predict(w,xb)
  error = output-y
  m = y.shape[0]
  #rergularization
  reg = (lamda / m) * w
  reg[0] = 0
  gradient = (xb.T @ error) / m
  w = w - alpha*(gradient+reg)
  return w

def train(X, y, X_val, y_val, alpha=0.1, iters=500, verbose=True,
lamda=0.0):
    np.random.seed(RANDSEED)
    w = np.random.randn(X.shape[1]) * 0.01
    history_train = []
    history_val = []

    for i in range(iters):
        pred = predict(w, X)
        loss = cost(y, pred, w, lamda)
        history_train.append(loss)

        pred_val = predict(w, X_val)
        loss_val = cost(y_val, pred_val)
        history_val.append(loss_val)
        if verbose:
          print("Iterasi ke-",i,"  Train Loss: ",loss, "  Val
Loss:",loss_val)
        w = update_bobot(w, X, y, alpha, lamda)

    return w, np.array(history_train), np.array(history_val)

def train_test_split(X, y, test_ratio=0.2, seed=RANDSEED):
    rng = np.random.default_rng(seed)
    idx = np.arange(len(y))
    rng.shuffle(idx)
    n_test = int(len(y) * test_ratio)
    test_idx = idx[:n_test]
    train_idx = idx[n_test:]
    return X[train_idx], X[test_idx], y[train_idx], y[test_idx]

def pilih_sample_random(X, y, n, seed=RANDSEED):
    rng = np.random.default_rng(seed)
    idx = rng.choice(X.shape[0], n, replace=False)
    X_sample = X[idx]
    y_sample = y[idx]
    return X_sample, y_sample

def plot_history_train_validation(hist_train, hist_val):
    plt.plot(hist_train, label="Train Loss")
```

```
    plt.plot(hist_val, label="Validation Loss")
    plt.xlabel("Iterasi")
    plt.ylabel("Loss")
    plt.title("Training Curve")
    plt.legend()
    plt.show()

def mse(y_true, y_pred):
    return np.mean((y_true - y_pred)**2)
```

##1. Persipan Dataset

###a. Baca Dataset dan Lakukan Preview

```
#Baca Dataset
data = pd.read_csv("https://drive.google.com/uc?id=16g89ugB-pdYeloUQUhvWLxaIYpTi3e1A")
#Lihat Summary Dataset
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1500 entries, 0 to 1499
Data columns (total 13 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   luas_bangunan_m2            1500 non-null   float64
 1   jumlah_kamar_tidur          1500 non-null   int64
 2   jarak_ke_pusat_kota_km      1500 non-null   float64
 3   usia_rumah_tahun            1500 non-null   float64
 4   tinggi_pagar_cm             1500 non-null   float64
 5   umur_cat_garasi_tahun       1500 non-null   float64
 6   panjang_gorden_m            1500 non-null   float64
 7   jumlah_tanaman_hias         1500 non-null   int64
 8   jarak_ke_tiang_listrik_m    1500 non-null   float64
 9   nomor_rumah_digit_terakhir  1500 non-null   int64
 10  jumlah_dekorasi_dinding     1500 non-null   int64
 11  tinggi_mailbox_cm           1500 non-null   float64
 12  harga_juta_rupiah           1500 non-null   float64
dtypes: float64(9), int64(4)
memory usage: 152.5 KB
```

```
#Lihat statistik setiap fitur
data.describe()
```

```
       luas_bangunan_m2  jumlah_kamar_tidur  jarak_ke_pusat_kota_km  \
count       1500.000000         1500.000000             1500.000000
mean         109.467493            2.970667                9.126580
std           27.866111            1.401112                3.004256
min           36.000000            1.000000                1.000000
25%           90.990000            2.000000                7.067500
```

```
50%          110.100000            3.000000                 9.120000
75%          127.387500            4.000000                11.202500
max          199.010000            5.000000                18.270000

       usia_rumah_tahun  tinggi_pagar_cm  umur_cat_garasi_tahun  \
count       1500.000000      1500.000000            1500.000000
mean          11.823460       165.910840               3.980660
std            6.695807        18.412331               1.929571
min            0.000000       110.000000               0.000000
25%            6.937500       153.735000               2.640000
50%           11.635000       165.430000               3.900000
75%           16.070000       178.512500               5.270000
max           36.180000       223.340000              12.000000

       panjang_gorden_m  jumlah_tanaman_hias  jarak_ke_tiang_listrik_m
\
count       1500.000000           1500.00000               1500.000000

mean           1.796220              7.02200                 17.966860

std            0.407312              4.29315                  6.001637

min            0.800000              0.00000                  2.000000

25%            1.530000              3.00000                 13.940000

50%            1.790000              7.00000                 17.975000

75%            2.070000             11.00000                 21.972500

max            3.170000             14.00000                 40.000000


       nomor_rumah_digit_terakhir  jumlah_dekorasi_dinding
tinggi_mailbox_cm  \
count                 1500.000000              1500.000000
1500.000000
mean                     4.513333                 5.363333
118.635980
std                      2.822375                 3.529253
14.059767
min                      0.000000                 0.000000
70.000000
25%                      2.000000                 2.000000
109.360000
50%                      5.000000                 5.000000
118.295000
75%                      7.000000                 8.000000
128.007500
max                      9.000000                11.000000
```

```
161.410000

       harga_juta_rupiah
count        1500.000000
mean          469.599053
std           100.873597
min           159.060000
25%           400.972500
50%           468.700000
75%           536.170000
max           773.880000
```

####PERTANYAAN 1

1. Berapa jumlah data dalam dataset?
2. Jika masalah yang akan diselesaikan adalah prediksi harga rumah, apa nama kolom labelnya?
3. Berapa jumlah fiturnya (tidak termasuk kolom label)?
4. Apakah range setiap fitur sama?

JAWABAN 1 (Jawaban Singkat Saja):

1. 1500
2. harga_juta_rupiah
3. 12
4. Tidak, range setiap fitur bervariasi bergantung min dan max kolom tersebut.

###b. Pisahkan Input dan Labelnya

```
#Pisahkan fitur input dan labelnya
X = data.drop(columns=["harga_juta_rupiah"]).to_numpy()
y = data["harga_juta_rupiah"].to_numpy()
#Lihat Shape dari X dan Y
print(X.shape)
print(y.shape)

(1500, 12)
(1500,)
```

###c. Lakukan Scaling

Lakukan scaling menggunakan Min-Max Scaling.

$ x' = \frac{x - \min(x)}{\max(x) - \min(x)} $

```
X_min = X.min(axis=0)
X_max = X.max(axis=0)
print("X_min sebelum scaling:", X_min)
print("X_max sebelum scaling:", X_max)
```

```
#Lakukan Scaling
X = (X - X_min) / (X_max - X_min)

X_min = X.min(axis=0)
X_max = X.max(axis=0)
print("X_min setelah scaling:", X_min)
print("X_max setelah scaling:", X_max)
```

```
X_min sebelum scaling: [ 36.    1.    1.    0.  110.    0.    0.8   0.
2.    0.    0.   70. ]
X_max sebelum scaling: [199.01   5.    18.27  36.18 223.34  12.
3.17  14.    40.     9.
  11.   161.41]
X_min setelah scaling: [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
X_max setelah scaling: [1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1.]
```

####PERTANYAAN 2

1. Kenapa kita perlu melakukan Scaling?
2. Apa dampaknya jika kita tidak melakukan scaling?

JAWABAN 2:

1. Agar semua fitur memiliki rentang nilai yang seragam, sehingga model konvergen lebih cepat dan fitur dengan rentang besar tidak mandominasi model.
2. Proses training (konvergensi) akan lama dan bisa jadi model tidak mendapatkan bobot yang optimal karena error didominasi fitur berentang besar.

###d. Tambahkan bias pada X

```
print('Shape sebelum ditambah bias: ',X.shape)
X = add_bias(X)
print('Shape setelah ditambah bias: ',X.shape)
```

```
Shape sebelum ditambah bias:  (1500, 12)
Shape setelah ditambah bias:  (1500, 13)
```

###e. Lakukan Pemisahan/Split Dataset Menjadi Data Training dan Testing

```
X_train, X_test, y_train, y_test = train_test_split(X,y,0.2)
print("Shape Data Training: ", X_train.shape)
print("Shape Data Testing: ", X_test.shape)
print("Shape Label Training: ", y_train.shape)
print("Shape Label Training: ", y_test.shape)
```

```
Shape Data Training:  (1200, 13)
Shape Data Testing:  (300, 13)
Shape Label Training:  (1200,)
Shape Label Training:  (300,)
```

####PERTANYAAN 3

1. Kenapa kita perlu melakukan split dataset menjaid training dan testing?
2. Berapa jumlah data training?
3. Berapa jumlah data testing?
4. Berapa rasio data testing dibanding jml data keseluruhan?

JAWABAN 3:

1. Untuk mengevaluasi performa dan kemampuan generalisasi model pada data baru yang belum pernah dilihat sebelumnya.
2. 1200
3. 300
4. 0.2 (20%)

##2. Training Model

###a. Pisahkan Data Training Sungguhan dan Data Validation

```
X_train_sungguhan, X_val, y_train_sungguhan, y_val =
train_test_split(X_train,y_train,0.2)
print("Shape data training: ", X_train_sungguhan.shape)
print("Shape data validation: ", X_val.shape)
```

```
Shape data training:  (960, 13)
Shape data validation:  (240, 13)
```

###b. Training Menggunakan Seluruh Data Training

```
#Training model
w, hist_train, hist_val = train(X_train_sungguhan, y_train_sungguhan,
X_val, y_val, 0.1, 500)

#Plot History Train dan Validation Loss
plot_history_train_validation(hist_train, hist_val)
```

```
Iterasi ke- 0   Train Loss:  115489.74437607863   Val Loss:
115401.82447268326
Iterasi ke- 1   Train Loss:  50849.356629945774   Val Loss:
50500.924015164775
Iterasi ke- 2   Train Loss:  23879.412228210844   Val Loss:
23417.91068500398
Iterasi ke- 3   Train Loss:  12611.957968104523   Val Loss:
12101.466744415935
Iterasi ke- 4   Train Loss:  7890.129249618846   Val Loss:
7358.8501112002905
Iterasi ke- 5   Train Loss:  5897.081291055164   Val Loss:
5357.605987153432
Iterasi ke- 6   Train Loss:  5041.838932763789   Val Loss:
4499.876186235266
```

```
Iterasi ke- 7   Train Loss:  4661.231829791513   Val Loss:
4119.4080421560375
Iterasi ke- 8   Train Loss:  4478.824771540735   Val Loss:
3938.3626904645953
Iterasi ke- 9   Train Loss:  4379.384718985007   Val Loss:
3840.857220275837
Iterasi ke- 10   Train Loss:  4314.872361149979   Val Loss:
3778.5471647397985
Iterasi ke- 11   Train Loss:  4265.256765775975   Val Loss:
3731.2570356860224
Iterasi ke- 12   Train Loss:  4222.181827341902   Val Loss:
3690.5622589583973
Iterasi ke- 13   Train Loss:  4182.157982736796   Val Loss:
3652.9412942618364
Iterasi ke- 14   Train Loss:  4143.724324682828   Val Loss:
3616.9188348809353
Iterasi ke- 15   Train Loss:  4106.2656879460865   Val Loss:
3581.8737165083994
Iterasi ke- 16   Train Loss:  4069.5195745211972   Val Loss:
3547.541348336315
Iterasi ke- 17   Train Loss:  4033.370623050846   Val Loss:
3513.8059833767857
Iterasi ke- 18   Train Loss:  3997.7649347920146   Val Loss:
3480.6139847602667
Iterasi ke- 19   Train Loss:  3962.674359213945   Val Loss:
3447.9376201145496
Iterasi ke- 20   Train Loss:  3928.081604239117   Val Loss:
3415.7599622965763
Iterasi ke- 21   Train Loss:  3893.974026848248   Val Loss:
3384.068606888157
Iterasi ke- 22   Train Loss:  3860.341041972865   Val Loss:
3352.85307077189
Iterasi ke- 23   Train Loss:  3827.173039658903   Val Loss:
3322.103724969339
Iterasi ke- 24   Train Loss:  3794.4609309858647   Val Loss:
3291.811364396885
Iterasi ke- 25   Train Loss:  3762.1959561333565   Val Loss:
3261.967039822162
Iterasi ke- 26   Train Loss:  3730.3696017809175   Val Loss:
3232.561995981234
Iterasi ke- 27   Train Loss:  3698.973564142468   Val Loss:
3203.5876512891564
Iterasi ke- 28   Train Loss:  3667.9997310825274   Val Loss:
3175.035592775453
Iterasi ke- 29   Train Loss:  3637.440172244783   Val Loss:
3146.8975757570765
Iterasi ke- 30   Train Loss:  3607.2871325775895   Val Loss:
3119.165524298844
Iterasi ke- 31   Train Loss:  3577.533027331199   Val Loss:
```

```
3091.8315311480974
Iterasi ke- 32   Train Loss:  3548.17043772289   Val Loss:
3064.8878568499804
Iterasi ke- 33   Train Loss:  3519.192106933671   Val Loss:
3038.3269281092876
Iterasi ke- 34   Train Loss:  3490.5909362950765   Val Loss:
3012.1413355632912
Iterasi ke- 35   Train Loss:  3462.3599816058477   Val Loss:
2986.3238311321743
Iterasi ke- 36   Train Loss:  3434.4924495522146   Val Loss:
2960.8673250857614
Iterasi ke- 37   Train Loss:  3406.981694219612   Val Loss:
2935.7648829327622
Iterasi ke- 38   Train Loss:  3379.8212136896336   Val Loss:
2911.0097222101936
Iterasi ke- 39   Train Loss:  3353.0046467185125   Val Loss:
2886.5952092281796
Iterasi ke- 40   Train Loss:  3326.525769494471   Val Loss:
2862.514855808505
Iterasi ke- 41   Train Loss:  3300.3784924717866   Val Loss:
2838.762316043166
Iterasi ke- 42   Train Loss:  3274.556857279605   Val Loss:
2815.3313830904494
Iterasi ke- 43   Train Loss:  3249.055033703691   Val Loss:
2792.2159860201045
Iterasi ke- 44   Train Loss:  3223.867316739336   Val Loss:
2769.410186714883
Iterasi ke- 45   Train Loss:  3198.9881237137315   Val Loss:
2746.9081768329315
Iterasi ke- 46   Train Loss:  3174.4119914761536   Val Loss:
2724.7042748334848
Iterasi ke- 47   Train Loss:  3150.133573654331   Val Loss:
2702.792923067025
Iterasi ke- 48   Train Loss:  3126.147637975425   Val Loss:
2681.168684930163
Iterasi ke- 49   Train Loss:  3102.449063650091   Val Loss:
2659.8262420849073
Iterasi ke- 50   Train Loss:  3079.0328388181024   Val Loss:
2638.76039174161
Iterasi ke- 51   Train Loss:  3055.894058054095   Val Loss:
2617.966044004607
Iterasi ke- 52   Train Loss:  3033.027919931973   Val Loss:
2597.4382192794505
Iterasi ke- 53   Train Loss:  3010.4297246466117   Val Loss:
2577.172045740486
Iterasi ke- 54   Train Loss:  2988.0948716914677   Val Loss:
2557.162756857537
Iterasi ke- 55   Train Loss:  2966.01885759078   Val Loss:
2537.405688980379
```

```
Iterasi ke- 56   Train Loss:  2944.1972736850635   Val Loss:
2517.8962789797242
Iterasi ke- 57   Train Loss:  2922.6258039686245   Val Loss:
2498.630061943405
Iterasi ke- 58   Train Loss:  2901.3002229778563   Val Loss:
2479.6026689265063
Iterasi ke- 59   Train Loss:  2880.2163937291266   Val Loss:
2460.809824754173
Iterasi ke- 60   Train Loss:  2859.3702657050453   Val Loss:
2442.2473458758536
Iterasi ke- 61   Train Loss:  2838.7578728880035   Val Loss:
2423.911138269801
Iterasi ke- 62   Train Loss:  2818.3753318398226   Val Loss:
2405.7971953966107
Iterasi ke- 63   Train Loss:  2798.218839826445   Val Loss:
2387.9015962006683
Iterasi ke- 64   Train Loss:  2778.284672986579   Val Loss:
2370.2205031583608
Iterasi ke- 65   Train Loss:  2758.569184543268   Val Loss:
2352.7501603719525
Iterasi ke- 66   Train Loss:  2739.068803057349   Val Loss:
2335.486891708044
Iterasi ke- 67   Train Loss:  2719.7800307218263   Val Loss:
2318.427098979569
Iterasi ke- 68   Train Loss:  2700.699441696163   Val Loss:
2301.5672601702854
Iterasi ke- 69   Train Loss:  2681.8236804795715   Val Loss:
2284.903927700771
Iterasi ke- 70   Train Loss:  2663.1494603223505   Val Loss:
2268.4337267349356
Iterasi ke- 71   Train Loss:  2644.6735616743867   Val Loss:
2252.1533535260924
Iterasi ke- 72   Train Loss:  2626.3928306699268   Val Loss:
2236.0595738016555
Iterasi ke- 73   Train Loss:  2608.3041776477653   Val Loss:
2220.149221185552
Iterasi ke- 74   Train Loss:  2590.4045757060067   Val Loss:
2204.4191956574577
Iterasi ke- 75   Train Loss:  2572.6910592905797   Val Loss:
2188.866462047983
Iterasi ke- 76   Train Loss:  2555.1607228167104   Val Loss:
2173.488048568965
Iterasi ke- 77   Train Loss:  2537.810719322559   Val Loss:
2158.2810453780344
Iterasi ke- 78   Train Loss:  2520.6382591542842   Val Loss:
2143.242603176647
Iterasi ke- 79   Train Loss:  2503.6406086817506   Val Loss:
2128.369931840797
Iterasi ke- 80   Train Loss:  2486.815089044198   Val Loss:
```

```
2113.6602990836286
Iterasi ke- 81   Train Loss:  2470.159074925126   Val Loss:
2099.111029149203
Iterasi ke- 82   Train Loss:  2453.669993355724   Val Loss:
2084.7195015366856
Iterasi ke- 83   Train Loss:  2437.3453225461603   Val Loss:
2070.483149754228
Iterasi ke- 84   Train Loss:  2421.182590744079   Val Loss:
2056.3994601018567
Iterasi ke- 85   Train Loss:  2405.179375119648   Val Loss:
2042.4659704826665
Iterasi ke- 86   Train Loss:  2389.333300676546   Val Loss:
2028.680269241678
Iterasi ke- 87   Train Loss:  2373.6420391882625   Val Loss:
2015.039994031675
Iterasi ke- 88   Train Loss:  2358.103308159111   Val Loss:
2001.542830705415
Iterasi ke- 89   Train Loss:  2342.714869809388   Val Loss:
1988.186512233573
Iterasi ke- 90   Train Loss:  2327.474530084081   Val Loss:
1974.9688176478162
Iterasi ke- 91   Train Loss:  2312.380137684595   Val Loss:
1961.8875710084274
Iterasi ke- 92   Train Loss:  2297.429583122928   Val Loss:
1948.9406403958824
Iterasi ke- 93   Train Loss:  2282.620797797792   Val Loss:
1936.1259369258319
Iterasi ke- 94   Train Loss:  2267.9517530921353   Val Loss:
1923.4414137869287
Iterasi ke- 95   Train Loss:  2253.420459491575   Val Loss:
1910.885065300964
Iterasi ke- 96   Train Loss:  2239.0249657232516   Val Loss:
1898.454926004793
Iterasi ke- 97   Train Loss:  2224.7633579146013   Val Loss:
1886.1490697535255
Iterasi ke- 98   Train Loss:  2210.633758771599   Val Loss:
1873.9656088444967
Iterasi ke- 99   Train Loss:  2196.634326776   Val Loss:
1861.9026931615142
Iterasi ke- 100   Train Loss:  2182.7632554011343   Val Loss:
1849.9585093389148
Iterasi ke- 101   Train Loss:  2169.018772345823   Val Loss:
1838.1312799449624
Iterasi ke- 102   Train Loss:  2155.3991387859696   Val Loss:
1826.4192626841307
Iterasi ke- 103   Train Loss:  2141.902648643438   Val Loss:
1814.820749617826
Iterasi ke- 104   Train Loss:  2128.5276278717843   Val Loss:
1803.3340664031243
```

```
Iterasi ke- 105   Train Loss:  2115.2724337584614   Val Loss:
1791.9575715490862
Iterasi ke- 106   Train Loss:  2102.135454243097   Val Loss:
1780.6896556902536
Iterasi ke- 107   Train Loss:  2089.1151072514785   Val Loss:
1769.5287408769075
Iterasi ke- 108   Train Loss:  2076.209840044863   Val Loss:
1758.4732798817056
Iterasi ke- 109   Train Loss:  2063.4181285842556   Val Loss:
1747.5217555223105
Iterasi ke- 110   Train Loss:  2050.7384769093114   Val Loss:
1736.6726799996331
Iterasi ke- 111   Train Loss:  2038.1694165315043   Val Loss:
1725.9245942513264
Iterasi ke- 112   Train Loss:  2025.7095058412353   Val Loss:
1715.2760673201687
Iterasi ke- 113   Train Loss:  2013.3573295285512   Val Loss:
1704.7256957369925
Iterasi ke- 114   Train Loss:  2001.1114980171503   Val Loss:
1694.2721029178124
Iterasi ke- 115   Train Loss:  1988.9706469113678   Val Loss:
1683.9139385748229
Iterasi ke- 116   Train Loss:  1976.9334364558313   Val Loss:
1673.6498781409377
Iterasi ke- 117   Train Loss:  1964.9985510074907   Val Loss:
1663.4786222075572
Iterasi ke- 118   Train Loss:  1953.1646985197333   Val Loss:
1653.39889597525
Iterasi ke- 119   Train Loss:  1941.4306100382973   Val Loss:
1643.4094487170494
Iterasi ke- 120   Train Loss:  1929.7950392087087   Val Loss:
1633.5090532540685
Iterasi ke- 121   Train Loss:  1918.2567617949694   Val Loss:
1623.6965054431432
Iterasi ke- 122   Train Loss:  1906.8145752092373   Val Loss:
1613.970623676226
Iterasi ke- 123   Train Loss:  1895.467298052232   Val Loss:
1604.3302483912494
Iterasi ke- 124   Train Loss:  1884.2137696641232   Val Loss:
1594.774241594198
Iterasi ke- 125   Train Loss:  1873.052849685652   Val Loss:
1585.3014863921208
Iterasi ke- 126   Train Loss:  1861.983417629245   Val Loss:
1575.9108865368325
Iterasi ke- 127   Train Loss:  1851.0043724598868   Val Loss:
1566.6013659790476
Iterasi ke- 128   Train Loss:  1840.1146321855267   Val Loss:
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Iterasi ke- 129   Train Loss:  1829.3131334567888   Val Loss:
```

```
1548.221356949297
Iterasi ke- 130   Train Loss:  1818.5988311757771   Val Loss:
1539.1488135017828
Iterasi ke- 131   Train Loss:  1807.9706981137524   Val Loss:
1530.1532385781854
Iterasi ke- 132   Train Loss:  1797.4277245374822   Val Loss:
1521.23365078432
Iterasi ke- 133   Train Loss:  1786.9689178440576   Val Loss:
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Iterasi ke- 134   Train Loss:  1776.5933022039758   Val Loss:
1503.6185992779006
Iterasi ke- 135   Train Loss:  1766.2999182123021   Val Loss:
1494.9212599165046
Iterasi ke- 136   Train Loss:  1756.0878225477163   Val Loss:
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Iterasi ke- 137   Train Loss:  1745.9560876392648   Val Loss:
1477.74239003724
Iterasi ke- 138   Train Loss:  1735.9038013406312   Val Loss:
1469.259082529126
Iterasi ke- 139   Train Loss:  1725.9300666117572   Val Loss:
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Iterasi ke- 140   Train Loss:  1716.0340012076426   Val Loss:
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Iterasi ke- 141   Train Loss:  1706.2147373741468   Val Loss:
1444.2233350089812
Iterasi ke- 142   Train Loss:  1696.4714215506413   Val Loss:
1436.0133612475672
Iterasi ke- 143   Train Loss:  1686.8032140793453   Val Loss:
1427.8696701061408
Iterasi ke- 144   Train Loss:  1677.2092889211942   Val Loss:
1419.7914697154597
Iterasi ke- 145   Train Loss:  1667.6888333780828   Val Loss:
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Iterasi ke- 146   Train Loss:  1658.2410478213385   Val Loss:
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Iterasi ke- 147   Train Loss:  1648.8651454262836   Val Loss:
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Iterasi ke- 148   Train Loss:  1639.5603519127321   Val Loss:
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Iterasi ke- 149   Train Loss:  1630.3259052913002   Val Loss:
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Iterasi ke- 150   Train Loss:  1621.161055615382   Val Loss:
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Iterasi ke- 151   Train Loss:  1612.0650647386638   Val Loss:
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Iterasi ke- 152   Train Loss:  1603.0372060780492   Val Loss:
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Iterasi ke- 153   Train Loss:  1594.0767643818674   Val Loss:
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Iterasi ke- 154   Train Loss:  1585.1830355032437   Val Loss:
1342.4466858989642
Iterasi ke- 155   Train Loss:  1576.3553261785116   Val Loss:
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Iterasi ke- 156   Train Loss:  1567.5929538105481   Val Loss:
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Iterasi ke- 157   Train Loss:  1558.8952462569223   Val Loss:
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Iterasi ke- 158   Train Loss:  1550.2615416227418   Val Loss:
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Iterasi ke- 159   Train Loss:  1541.6911880580897   Val Loss:
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Iterasi ke- 160   Train Loss:  1533.1835435599457   Val Loss:
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Iterasi ke- 161   Train Loss:  1524.7379757784895   Val Loss:
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Iterasi ke- 162   Train Loss:  1516.3538618276784   Val Loss:
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Iterasi ke- 163   Train Loss:  1508.030588100013   Val Loss:
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Iterasi ke- 164   Train Loss:  1499.7675500853793   Val Loss:
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Iterasi ke- 165   Train Loss:  1491.564152193883   Val Loss:
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Iterasi ke- 166   Train Loss:  1483.4198075825846   Val Loss:
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Iterasi ke- 167   Train Loss:  1475.3339379860336   Val Loss:
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Iterasi ke- 168   Train Loss:  1467.3059735505312   Val Loss:
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Iterasi ke- 169   Train Loss:  1459.3353526720223   Val Loss:
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Iterasi ke- 170   Train Loss:  1451.4215218375373   Val Loss:
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Iterasi ke- 171   Train Loss:  1443.5639354701052   Val Loss:
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Iterasi ke- 172   Train Loss:  1435.762055777051   Val Loss:
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Iterasi ke- 173   Train Loss:  1428.0153526016104   Val Loss:
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Iterasi ke- 174   Train Loss:  1420.32330327777   Val Loss:
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Iterasi ke- 175   Train Loss:  1412.6853924882773   Val Loss:
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Iterasi ke- 176   Train Loss:  1405.1011121257318   Val Loss:
1191.7634174518066
Iterasi ke- 177   Train Loss:  1397.5699611566952   Val Loss:
1185.4786909483994
Iterasi ke- 178   Train Loss:  1390.0914454887495   Val Loss:
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```
1179.239099011219
Iterasi ke- 179   Train Loss:  1382.6650778404335   Val Loss:
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Iterasi ke- 180   Train Loss:  1375.2903776139965   Val Loss:
1166.893534406174
Iterasi ke- 181   Train Loss:  1367.9668707708968   Val Loss:
1160.7866860753861
Iterasi ke- 182   Train Loss:  1360.694089709992   Val Loss:
1154.7232207265572
Iterasi ke- 183   Train Loss:  1353.471573148351   Val Loss:
1148.7027165034976
Iterasi ke- 184   Train Loss:  1346.2988660046308   Val Loss:
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Iterasi ke- 185   Train Loss:  1339.1755192849635   Val Loss:
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Iterasi ke- 186   Train Loss:  1332.101089971289   Val Loss:
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Iterasi ke- 187   Train Loss:  1325.0751409120828   Val Loss:
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Iterasi ke- 188   Train Loss:  1318.0972407154215   Val Loss:
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Iterasi ke- 189   Train Loss:  1311.166963644336   Val Loss:
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Iterasi ke- 190   Train Loss:  1304.2838895143932   Val Loss:
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Iterasi ke- 191   Train Loss:  1297.4476035934683   Val Loss:
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Iterasi ke- 192   Train Loss:  1290.6576965036413   Val Loss:
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Iterasi ke- 193   Train Loss:  1283.9137641251848   Val Loss:
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Iterasi ke- 194   Train Loss:  1277.215407502584   Val Loss:
1085.1978953197156
Iterasi ke- 195   Train Loss:  1270.562232752551   Val Loss:
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Iterasi ke- 196   Train Loss:  1263.9538509739823   Val Loss:
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Iterasi ke- 197   Train Loss:  1257.389878159819   Val Loss:
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Iterasi ke- 198   Train Loss:  1250.8699351107657   Val Loss:
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Iterasi ke- 199   Train Loss:  1244.3936473508245   Val Loss:
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Iterasi ke- 200   Train Loss:  1237.9606450446042   Val Loss:
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Iterasi ke- 201   Train Loss:  1231.5705629163674   Val Loss:
1047.2336911821542
Iterasi ke- 202   Train Loss:  1225.2230401707698   Val Loss:
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```

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Iterasi ke- 203   Train Loss:  1218.9177204152575   Val Loss:
1036.715649322295
Iterasi ke- 204   Train Loss:  1212.654251584088   Val Loss:
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Iterasi ke- 205   Train Loss:  1206.43228586393   Val Loss:
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Iterasi ke- 206   Train Loss:  1200.2514796210164   Val Loss:
1021.2030241392536
Iterasi ke- 207   Train Loss:  1194.1114933298034   Val Loss:
1016.1014371938476
Iterasi ke- 208   Train Loss:  1188.0119915031148   Val Loss:
1011.0339940233006
Iterasi ke- 209   Train Loss:  1181.9526426237323   Val Loss:
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Iterasi ke- 210   Train Loss:  1175.9331190773921   Val Loss:
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Iterasi ke- 211   Train Loss:  1169.9530970871706   Val Loss:
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Iterasi ke- 212   Train Loss:  1164.012256649218   Val Loss:
991.0998279959243
Iterasi ke- 213   Train Loss:  1158.1102814698104   Val Loss:
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Iterasi ke- 214   Train Loss:  1152.2468589036957   Val Loss:
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Iterasi ke- 215   Train Loss:  1146.4216798936998   Val Loss:
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Iterasi ke- 216   Train Loss:  1140.6344389115663   Val Loss:
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Iterasi ke- 217   Train Loss:  1134.8848339000033   Val Loss:
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Iterasi ke- 218   Train Loss:  1129.1725662159079   Val Loss:
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Iterasi ke- 219   Train Loss:  1123.497340574745   Val Loss:
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Iterasi ke- 220   Train Loss:  1117.858864996055   Val Loss:
952.7843869941122
Iterasi ke- 221   Train Loss:  1112.2568507500566   Val Loss:
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Iterasi ke- 222   Train Loss:  1106.6910123053365   Val Loss:
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Iterasi ke- 223   Train Loss:  1101.1610672775867   Val Loss:
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Iterasi ke- 224   Train Loss:  1095.666736379374   Val Loss:
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Iterasi ke- 225   Train Loss:  1090.2077433709173   Val Loss:
929.8400915620991
Iterasi ke- 226   Train Loss:  1084.783815011848   Val Loss:
925.3403090789835
Iterasi ke- 227   Train Loss:  1079.394681013937   Val Loss:
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Iterasi ke- 228   Train Loss:  1074.0400739947604   Val Loss:
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Iterasi ke- 229   Train Loss:  1068.7197294322884   Val Loss:
912.014845578955
Iterasi ke- 230   Train Loss:  1063.4333856203757   Val Loss:
907.6302180350443
Iterasi ke- 231   Train Loss:  1058.1807836251296   Val Loss:
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Iterasi ke- 232   Train Loss:  1052.9616672421437   Val Loss:
898.9453992432938
Iterasi ke- 233   Train Loss:  1047.7757829545715   Val Loss:
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Iterasi ke- 234   Train Loss:  1042.622879892025   Val Loss:
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Iterasi ke- 235   Train Loss:  1037.5027097902807   Val Loss:
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Iterasi ke- 236   Train Loss:  1032.4150269517697   Val Loss:
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Iterasi ke- 237   Train Loss:  1027.3595882068444   Val Loss:
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Iterasi ke- 239   Train Loss:  1017.3444827316089   Val Loss:
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Iterasi ke- 240   Train Loss:  1012.3843419634414   Val Loss:
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Iterasi ke- 241   Train Loss:  1007.4554971408055   Val Loss:
861.2140851960272
Iterasi ke- 242   Train Loss:  1002.5577171784428   Val Loss:
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Iterasi ke- 243   Train Loss:  997.6907733018726   Val Loss:
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Iterasi ke- 244   Train Loss:  992.8544390136025   Val Loss:
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Iterasi ke- 245   Train Loss:  988.0484900599843   Val Loss:
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Iterasi ke- 246   Train Loss:  983.2727043987037   Val Loss:
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Iterasi ke- 247   Train Loss:  978.5268621668855   Val Loss:
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Iterasi ke- 248   Train Loss:  973.8107456498078   Val Loss:
833.3262381865924
Iterasi ke- 249   Train Loss:  969.1241392502063   Val Loss:
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Iterasi ke- 250   Train Loss:  964.4668294581589   Val Loss:
825.5822252763259
Iterasi ke- 251   Train Loss:  959.8386048215357   Val Loss:
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Iterasi ke- 252   Train Loss:  955.2392559170061   Val Loss:
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```
817.9350408773072
Iterasi ke- 253   Train Loss:  950.6685753215859   Val Loss:
814.1473130538802
Iterasi ke- 254   Train Loss:  946.1263575847186   Val Loss:
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Iterasi ke- 255   Train Loss:  941.6123992008736   Val Loss:
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Iterasi ke- 256   Train Loss:  937.1264985826541   Val Loss:
802.9254882487813
Iterasi ke- 257   Train Loss:  932.6684560344033   Val Loss:
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Iterasi ke- 258   Train Loss:  928.2380737262974   Val Loss:
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Iterasi ke- 259   Train Loss:  923.8351556689141   Val Loss:
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Iterasi ke- 260   Train Loss:  919.4595076882683   Val Loss:
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Iterasi ke- 261   Train Loss:  915.1109374013063   Val Loss:
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Iterasi ke- 262   Train Loss:  910.7892541918435   Val Loss:
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Iterasi ke- 263   Train Loss:  906.4942691869418   Val Loss:
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Iterasi ke- 264   Train Loss:  902.225795233715   Val Loss:
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Iterasi ke- 265   Train Loss:  897.9836468765516   Val Loss:
770.4928249259104
Iterasi ke- 266   Train Loss:  893.76764033475   Val Loss:
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Iterasi ke- 267   Train Loss:  889.5775934805538   Val Loss:
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Iterasi ke- 268   Train Loss:  885.4133258175813   Val Loss:
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Iterasi ke- 269   Train Loss:  881.2746584596383   Val Loss:
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753.2417620844398
Iterasi ke- 271   Train Loss:  873.073417040517   Val Loss:
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Iterasi ke- 272   Train Loss:  869.0104930724417   Val Loss:
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Iterasi ke- 273   Train Loss:  864.9724695557961   Val Loss:
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Iterasi ke- 274   Train Loss:  860.9591753504458   Val Loss:
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Iterasi ke- 275   Train Loss:  856.970440806968   Val Loss:
736.51442276854
Iterasi ke- 276   Train Loss:  853.0060977479442   Val Loss:
733.2301990717802
```

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Iterasi ke- 277   Train Loss:  849.0659794495732   Val Loss:
729.9660595096836
Iterasi ke- 278   Train Loss:  845.1499206236091   Val Loss:
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Iterasi ke- 279   Train Loss:  841.2577573996056   Val Loss:
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Iterasi ke- 280   Train Loss:  837.3893273074682   Val Loss:
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Iterasi ke- 281   Train Loss:  833.5444692603045   Val Loss:
717.1075794449049
Iterasi ke- 282   Train Loss:  829.723023537568   Val Loss:
713.9417990313137
Iterasi ke- 283   Train Loss:  825.9248317684873   Val Loss:
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Iterasi ke- 284   Train Loss:  822.1497369157786   Val Loss:
707.6679107494004
Iterasi ke- 285   Train Loss:  818.3975832596308   Val Loss:
704.5595404708489
Iterasi ke- 286   Train Loss:  814.668216381961   Val Loss:
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Iterasi ke- 287   Train Loss:  810.9614831509351   Val Loss:
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Iterasi ke- 288   Train Loss:  807.2772317057471   Val Loss:
695.3471747282172
Iterasi ke- 289   Train Loss:  803.6153114416485   Val Loss:
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Iterasi ke- 290   Train Loss:  799.9755729952302   Val Loss:
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Iterasi ke- 291   Train Loss:  796.3578682299461   Val Loss:
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Iterasi ke- 292   Train Loss:  792.7620502218736   Val Loss:
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Iterasi ke- 293   Train Loss:  789.1879732457086   Val Loss:
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Iterasi ke- 294   Train Loss:  785.635492760991   Val Loss:
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Iterasi ke- 295   Train Loss:  782.1044653985522   Val Loss:
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Iterasi ke- 296   Train Loss:  778.5947489471812   Val Loss:
671.5856051785163
Iterasi ke- 297   Train Loss:  775.10620234051   Val Loss:
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Iterasi ke- 298   Train Loss:  771.6386856441069   Val Loss:
665.8228378506187
Iterasi ke- 299   Train Loss:  768.1920600427776   Val Loss:
662.9674486924666
Iterasi ke- 300   Train Loss:  764.7661878280679   Val Loss:
660.1292357445487
Iterasi ke- 301   Train Loss:  761.3609323859672   Val Loss:
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```
657.3080851024126
Iterasi ke- 302   Train Loss:  757.9761581848051   Val Loss:
654.5038838004901
Iterasi ke- 303   Train Loss:  754.6117307633384   Val Loss:
651.7165198008875
Iterasi ke- 304   Train Loss:  751.2675167190268   Val Loss:
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Iterasi ke- 305   Train Loss:  747.9433836964917   Val Loss:
646.1918601294674
Iterasi ke- 306   Train Loss:  744.6392003761533   Val Loss:
643.4543449219382
Iterasi ke- 307   Train Loss:  741.3548364630457   Val Loss:
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Iterasi ke- 308   Train Loss:  738.0901626758044   Val Loss:
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Iterasi ke- 309   Train Loss:  734.8450507358226   Val Loss:
635.3397591773332
Iterasi ke- 310   Train Loss:  731.619373356574   Val Loss:
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Iterasi ke- 311   Train Loss:  728.4130042330996   Val Loss:
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Iterasi ke- 312   Train Loss:  725.225818031652   Val Loss:
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Iterasi ke- 313   Train Loss:  722.0576903794998   Val Loss:
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Iterasi ke- 314   Train Loss:  718.9084978548814   Val Loss:
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Iterasi ke- 315   Train Loss:  715.7781179771131   Val Loss:
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Iterasi ke- 316   Train Loss:  712.6664291968439   Val Loss:
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Iterasi ke- 317   Train Loss:  709.5733108864544   Val Loss:
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Iterasi ke- 318   Train Loss:  706.498643330599   Val Loss:
611.8530656841701
Iterasi ke- 319   Train Loss:  703.442307716887   Val Loss:
609.32057768968
Iterasi ke- 320   Train Loss:  700.4041861267013   Val Loss:
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Iterasi ke- 321   Train Loss:  697.3841615261507   Val Loss:
604.3006984144042
Iterasi ke- 322   Train Loss:  694.3821177571537   Val Loss:
601.8131137642711
Iterasi ke- 323   Train Loss:  691.3979395286527   Val Loss:
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Iterasi ke- 324   Train Loss:  688.4315124079537   Val Loss:
596.8821765677525
Iterasi ke- 325   Train Loss:  685.4827228121916   Val Loss:
594.4386350603869
```

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Iterasi ke- 326   Train Loss:  682.5514579999166   Val Loss:
592.0095868367949
Iterasi ke- 327   Train Loss:  679.6376060628014   Val Loss:
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Iterasi ke- 328   Train Loss:  676.7410559174648   Val Loss:
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Iterasi ke- 329   Train Loss:  673.8616972974118   Val Loss:
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Iterasi ke- 330   Train Loss:  670.9994207450865   Val Loss:
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Iterasi ke- 331   Train Loss:  668.1541176040366   Val Loss:
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Iterasi ke- 332   Train Loss:  665.3256800111867   Val Loss:
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Iterasi ke- 333   Train Loss:  662.5140008892182   Val Loss:
575.4043620512556
Iterasi ke- 334   Train Loss:  659.7189739390553   Val Loss:
573.087980943187
Iterasi ke- 335   Train Loss:  656.9404936324536   Val Loss:
570.7852834585542
Iterasi ke- 336   Train Loss:  654.1784552046918   Val Loss:
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Iterasi ke- 337   Train Loss:  651.4327546473588   Val Loss:
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Iterasi ke- 338   Train Loss:  648.7032887012446   Val Loss:
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Iterasi ke- 339   Train Loss:  645.9899548493211   Val Loss:
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Iterasi ke- 340   Train Loss:  643.2926513098209   Val Loss:
559.4740398616033
Iterasi ke- 341   Train Loss:  640.6112770294077   Val Loss:
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Iterasi ke- 342   Train Loss:  637.9457316764381   Val Loss:
555.0423435819976
Iterasi ke- 343   Train Loss:  635.2959156343109   Val Loss:
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Iterasi ke- 344   Train Loss:  632.661729994906   Val Loss:
550.6626796695278
Iterasi ke- 345   Train Loss:  630.043076552108   Val Loss:
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Iterasi ke- 346   Train Loss:  627.4398577954177   Val Loss:
546.3343967191303
Iterasi ke- 347   Train Loss:  624.8519769036403   Val Loss:
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Iterasi ke- 348   Train Loss:  622.279337738662   Val Loss:
542.056852665531
Iterasi ke- 349   Train Loss:  619.7218448393036   Val Loss:
539.9369095894813
Iterasi ke- 350   Train Loss:  617.1794034152522   Val Loss:
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```
537.8294146048706
Iterasi ke- 351   Train Loss:  614.6519193410725   Val Loss:
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Iterasi ke- 352   Train Loss:  612.1392991502934   Val Loss:
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Iterasi ke- 353   Train Loss:  609.6414500295682   Val Loss:
531.5808438508359
Iterasi ke- 354   Train Loss:  607.1582798129098   Val Loss:
529.5223696184753
Iterasi ke- 355   Train Loss:  604.6896969759985   Val Loss:
527.4759603883228
Iterasi ke- 356   Train Loss:  602.235610630559   Val Loss:
525.4415411554336
Iterasi ke- 357   Train Loss:  599.7959305188089   Val Loss:
523.4190374409434
Iterasi ke- 358   Train Loss:  597.3705670079747   Val Loss:
521.4083752872546
Iterasi ke- 359   Train Loss:  594.9594310848757   Val Loss:
519.4094812532827
Iterasi ke- 360   Train Loss:  592.5624343505752   Val Loss:
517.4222824097685
Iterasi ke- 361   Train Loss:  590.1794890150946   Val Loss:
515.4467063346523
Iterasi ke- 362   Train Loss:  587.8105078921941   Val Loss:
513.4826811085069
Iterasi ke- 363   Train Loss:  585.4554043942146   Val Loss:
511.53013531003563
Iterasi ke- 364   Train Loss:  583.1140925269824   Val Loss:
509.58899801162426
Iterasi ke- 365   Train Loss:  580.7864868847755   Val Loss:
507.6591987749549
Iterasi ke- 366   Train Loss:  578.4725026453478   Val Loss:
505.7406676466735
Iterasi ke- 367   Train Loss:  576.1720555650154   Val Loss:
503.83333515411636
Iterasi ke- 368   Train Loss:  573.8850619737971   Val Loss:
501.9371323010898
Iterasi ke- 369   Train Loss:  571.6114387706156   Val Loss:
500.05199056370367
Iterasi ke- 370   Train Loss:  569.3511034185524   Val Loss:
498.17784188625956
Iterasi ke- 371   Train Loss:  567.1039739401584   Val Loss:
496.3146186771891
Iterasi ke- 372   Train Loss:  564.8699689128199   Val Loss:
494.4622538050443
Iterasi ke- 373   Train Loss:  562.6490074641767   Val Loss:
492.62068059453935
Iterasi ke- 374   Train Loss:  560.4410092675935   Val Loss:
490.78983282263937
```

```
Iterasi ke- 375   Train Loss:  558.2458945376833   Val Loss:
488.9696447147008
Iterasi ke- 376   Train Loss:  556.0635840258798   Val Loss:
487.16005094065605
Iterasi ke- 377   Train Loss:  553.8939990160635   Val Loss:
485.3609866112484
Iterasi ke- 378   Train Loss:  551.7370613202336   Val Loss:
483.57238727431053
Iterasi ke- 379   Train Loss:  549.5926932742302   Val Loss:
481.79418891108986
Iterasi ke- 380   Train Loss:  547.4608177335041   Val Loss:
480.02632793261836
Iterasi ke- 381   Train Loss:  545.3413580689329   Val Loss:
478.268741176124
Iterasi ke- 382   Train Loss:  543.2342381626846   Val Loss:
476.521365901489
Iterasi ke- 383   Train Loss:  541.1393824041262   Val Loss:
474.7841397877476
Iterasi ke- 384   Train Loss:  539.0567156857761   Val Loss:
473.05700092962593
Iterasi ke- 385   Train Loss:  536.9861633993033   Val Loss:
471.33988783412383
Iterasi ke- 386   Train Loss:  534.9276514315671   Val Loss:
469.63273941713504
Iterasi ke- 387   Train Loss:  532.881106160701   Val Loss:
467.9354950001097
Iterasi ke- 388   Train Loss:  530.8464544522404   Val Loss:
466.24809430675356
Iterasi ke- 389   Train Loss:  528.8236236552881   Val Loss:
464.5704774597647
Iterasi ke- 390   Train Loss:  526.8125415987246   Val Loss:
462.90258497761204
Iterasi ke- 391   Train Loss:  524.8131365874559   Val Loss:
461.24435777134596
Iterasi ke- 392   Train Loss:  522.8253373987033   Val Loss:
459.5957371414483
Iterasi ke- 393   Train Loss:  520.8490732783295   Val Loss:
457.9566647747175
Iterasi ke- 394   Train Loss:  518.8842739372056   Val Loss:
456.32708274118846
Iterasi ke- 395   Train Loss:  516.9308695476158   Val Loss:
454.7069334910887
Iterasi ke- 396   Train Loss:  514.9887907396984   Val Loss:
453.096159851826
Iterasi ke- 397   Train Loss:  513.057968597924   Val Loss:
451.49470502501185
Iterasi ke- 398   Train Loss:  511.1383346576102   Val Loss:
449.9025125835173
Iterasi ke- 399   Train Loss:  509.2298209014732   Val Loss:
```

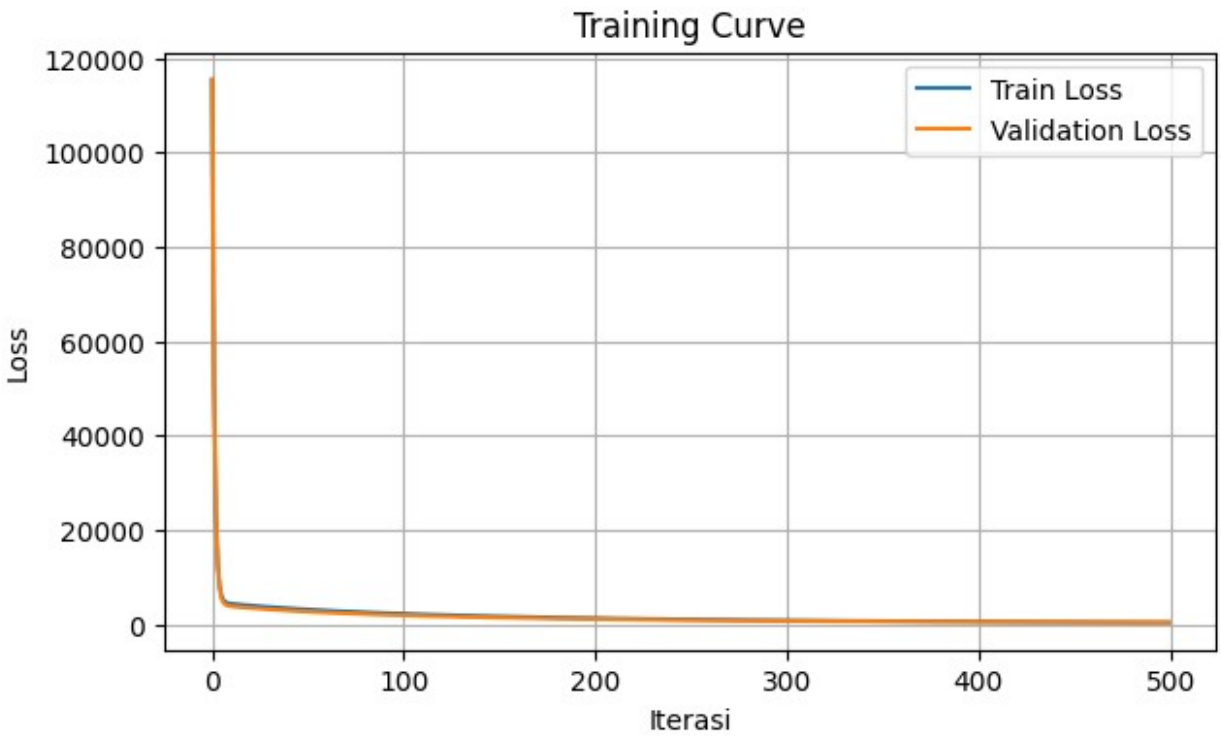
```
448.31952646856166
Iterasi ke- 400   Train Loss:  507.332359756212   Val Loss:
446.7456909868308
Iterasi ke- 401   Train Loss:  505.44588408913006   Val Loss:
445.18095080763203
Iterasi ke- 402   Train Loss:  503.57032720478946   Val Loss:
443.62525096007386
Iterasi ke- 403   Train Loss:  501.7056228416995   Val Loss:
442.07853683028236
Iterasi ke- 404   Train Loss:  499.8517051690395   Val Loss:
440.540754158641
Iterasi ke- 405   Train Loss:  498.0085087834112   Val Loss:
439.0118490370666
Iterasi ke- 406   Train Loss:  496.1759687056281   Val Loss:
437.4917679063104
Iterasi ke- 407   Train Loss:  494.35402037753266   Val Loss:
435.98045755328803
Iterasi ke- 408   Train Loss:  492.5425996588472   Val Loss:
434.4778651084393
Iterasi ke- 409   Train Loss:  490.74164282405593   Val Loss:
432.98393804311524
Iterasi ke- 410   Train Loss:  488.9510865593147   Val Loss:
431.49862416699136
Iterasi ke- 411   Train Loss:  487.1708679593955   Val Loss:
430.0218716255094
Iterasi ke- 412   Train Loss:  485.40092452465615   Val Loss:
428.55362889734454
Iterasi ke- 413   Train Loss:  483.6411941580428   Val Loss:
427.09384479190106
Iterasi ke- 414   Train Loss:  481.8916151621186   Val Loss:
425.6424684468282
Iterasi ke- 415   Train Loss:  480.15212623612376   Val Loss:
424.1994493255687
Iterasi ke- 416   Train Loss:  478.42266647306053   Val Loss:
422.764737214928
Iterasi ke- 417   Train Loss:  476.7031753568088   Val Loss:
421.3382822226674
Iterasi ke- 418   Train Loss:  474.9935927592674   Val Loss:
419.9200347751248
Iterasi ke- 419   Train Loss:  473.2938589375225   Val Loss:
418.5099456148562
Iterasi ke- 420   Train Loss:  471.6039145310442   Val Loss:
417.10796579830316
Iterasi ke- 421   Train Loss:  469.9237005589074   Val Loss:
415.7140466934817
Iterasi ke- 422   Train Loss:  468.2531584170392   Val Loss:
414.3281399776952
Iterasi ke- 423   Train Loss:  466.59222987549373   Val Loss:
412.95019763526955
```

```
Iterasi ke- 424   Train Loss:  464.94085707574925   Val Loss:
411.5801719553103
Iterasi ke- 425   Train Loss:  463.29898252803235   Val Loss:
410.2180155294822
Iterasi ke- 426   Train Loss:  461.6665491086666   Val Loss:
408.8636812498098
Iterasi ke- 427   Train Loss:  460.04350005744385   Val Loss:
407.51712230649963
Iterasi ke- 428   Train Loss:  458.42977897502254   Val Loss:
406.17829218578385
Iterasi ke- 429   Train Loss:  456.8253298203457   Val Loss:
404.8471446677839
Iterasi ke- 430   Train Loss:  455.23009690808703   Val Loss:
403.5236338243945
Iterasi ke- 431   Train Loss:  453.64402490611525   Val Loss:
402.2077140171888
Iterasi ke- 432   Train Loss:  452.06705883298514   Val Loss:
400.8993398953422
Iterasi ke- 433   Train Loss:  450.4991440554491   Val Loss:
399.59846639357687
Iterasi ke- 434   Train Loss:  448.9402262859909   Val Loss:
398.3050487301237
Iterasi ke- 435   Train Loss:  447.3902515803826   Val Loss:
397.01904240470674
Iterasi ke- 436   Train Loss:  445.8491663352621   Val Loss:
395.74040319654176
Iterasi ke- 437   Train Loss:  444.3169172857324   Val Loss:
394.4690871623586
Iterasi ke- 438   Train Loss:  442.7934515029818   Val Loss:
393.2050506344367
Iterasi ke- 439   Train Loss:  441.2787163919256   Val Loss:
391.9482502186623
Iterasi ke- 440   Train Loss:  439.77265968886763   Val Loss:
390.6986427926025
Iterasi ke- 441   Train Loss:  438.2752294591833   Val Loss:
389.45618550359563
Iterasi ke- 442   Train Loss:  436.7863740950208   Val Loss:
388.2208357668613
Iterasi ke- 443   Train Loss:  435.30604231302476   Val Loss:
386.99255126362453
Iterasi ke- 444   Train Loss:  433.83418315207757   Val Loss:
385.77128993925993
Iterasi ke- 445   Train Loss:  432.37074597106056   Val Loss:
384.5570100014501
Iterasi ke- 446   Train Loss:  430.91568044663524   Val Loss:
383.3496699183626
Iterasi ke- 447   Train Loss:  429.46893657104215   Val Loss:
382.14922841684034
Iterasi ke- 448   Train Loss:  428.0304646499205   Val Loss:
```

```
380.95564448061134
Iterasi ke- 449   Train Loss:  426.60021530014296   Val Loss:
379.7688773485113
Iterasi ke- 450   Train Loss:  425.1781394476722   Val Loss:
378.5888865127243
Iterasi ke- 451   Train Loss:  423.76418832543436   Val Loss:
377.4156317170366
Iterasi ke- 452   Train Loss:  422.3583134712084   Val Loss:
376.2490729551079
Iterasi ke- 453   Train Loss:  420.96046672553706   Val Loss:
375.0891704687561
Iterasi ke- 454   Train Loss:  419.5706002296509   Val Loss:
373.93588474625653
Iterasi ke- 455   Train Loss:  418.1886664234136   Val Loss:
372.7891765206589
Iterasi ke- 456   Train Loss:  416.8146180432817   Val Loss:
371.64900676811374
Iterasi ke- 457   Train Loss:  415.4484081202818   Val Loss:
370.5153367062187
Iterasi ke- 458   Train Loss:  414.08998997800506   Val Loss:
369.3881277923738
Iterasi ke- 459   Train Loss:  412.73931723061764   Val Loss:
368.26734172215464
Iterasi ke- 460   Train Loss:  411.39634378088795   Val Loss:
367.1529404276971
Iterasi ke- 461   Train Loss:  410.0610238182293   Val Loss:
366.04488607609676
Iterasi ke- 462   Train Loss:  408.7333118167589   Val Loss:
364.9431410678209
Iterasi ke- 463   Train Loss:  407.41316253337317   Val Loss:
363.847668035135
Iterasi ke- 464   Train Loss:  406.1005310058375   Val Loss:
362.7584298405406
Iterasi ke- 465   Train Loss:  404.7953725508925   Val Loss:
361.6753895752284
Iterasi ke- 466   Train Loss:  403.49764276237477   Val Loss:
360.5985105575426
Iterasi ke- 467   Train Loss:  402.2072975093533   Val Loss:
359.527756331457
Iterasi ke- 468   Train Loss:  400.9242929342806   Val Loss:
358.46309066506785
Iterasi ke- 469   Train Loss:  399.64858545115845   Val Loss:
357.4044775490929
Iterasi ke- 470   Train Loss:  398.38013174371866   Val Loss:
356.35188119538805
Iterasi ke- 471   Train Loss:  397.1188887636174   Val Loss:
355.3052660354735
Iterasi ke- 472   Train Loss:  395.8648137286451   Val Loss:
354.2645967190726
```

```
Iterasi ke- 473   Train Loss:  394.6178641209499   Val Loss:
353.22983811266096
Iterasi ke- 474   Train Loss:  393.37799768527464   Val Loss:
352.200955298031
Iterasi ke- 475   Train Loss:  392.1451724272078   Val Loss:
351.1779135708631
Iterasi ke- 476   Train Loss:  390.91934661145   Val Loss:
350.1606784393129
Iterasi ke- 477   Train Loss:  389.70047876009085   Val Loss:
349.1492156226059
Iterasi ke- 478   Train Loss:  388.4885276509013   Val Loss:
348.1434910496459
Iterasi ke- 479   Train Loss:  387.2834523156401   Val Loss:
347.1434708576344
Iterasi ke- 480   Train Loss:  386.08521203837074   Val Loss:
346.14912139069827
Iterasi ke- 481   Train Loss:  384.8937663537935   Val Loss:
345.1604091985317
Iterasi ke- 482   Train Loss:  383.7090750455899   Val Loss:
344.17730103504726
Iterasi ke- 483   Train Loss:  382.53109814477915   Val Loss:
343.19976385703563
Iterasi ke- 484   Train Loss:  381.3597959280889   Val Loss:
342.2277648228404
Iterasi ke- 485   Train Loss:  380.1951289163361   Val Loss:
341.2612712910383
Iterasi ke- 486   Train Loss:  379.0370578728224   Val Loss:
340.3002508191331
Iterasi ke- 487   Train Loss:  377.88554380174065   Val Loss:
339.34467116225784
Iterasi ke- 488   Train Loss:  376.74054794659423   Val Loss:
338.3945002718871
Iterasi ke- 489   Train Loss:  375.6020317886272   Val Loss:
337.4497062945612
Iterasi ke- 490   Train Loss:  374.4699570452676   Val Loss:
336.5102575706176
Iterasi ke- 491   Train Loss:  373.34428566858213   Val Loss:
335.57612263293294
Iterasi ke- 492   Train Loss:  372.22497984374155   Val Loss:
334.6472702056755
Iterasi ke- 493   Train Loss:  371.1120019874997   Val Loss:
333.7236692030658
Iterasi ke- 494   Train Loss:  370.00531474668134   Val Loss:
332.80528872814796
Iterasi ke- 495   Train Loss:  368.90488099668363   Val Loss:
331.8920980715697
Iterasi ke- 496   Train Loss:  367.81066383998717   Val Loss:
330.9840667103706
Iterasi ke- 497   Train Loss:  366.7226266046793   Val Loss:
```

```
330.08116430678143
Iterasi ke- 498   Train Loss:  365.64073284298695   Val Loss:
329.1833607070302
Iterasi ke- 499   Train Loss:  364.5649463298229   Val Loss:
328.29062594016125
```



###c. Lakukan Evaluasi/Testing Menggunakan Data Testing

```
#Untuk Evaluasi Model, kita gunakan metric Mean Square Error (MSE)

#Evaluasi dengan data training
hasil_prediksi = predict(w,X_train_sungguhan)
error = mse(y_train_sungguhan, hasil_prediksi)
print("Error Trainig: ", error)

#Evaluasi dengan data TEsting
hasil_prediksi = predict(w,X_test)
error = mse(y_test, hasil_prediksi)
print("Error Testing: ", error)
```

```
Error Trainig:  726.9904621226783
Error Testing:  757.5708333814307
```

####PERTANYAAN 4

1. Berapa learning rate dan iterasi yang digunakan?
2. Berapa Error memorization dan generalization yang didapat?

3. Lakukan minimal 5 eksperimen untuk mendapatkan hasil terbaik, dengan mengubah jumlah iterasi dan learning rate, dokumentasikan hasilnya.
4. Berapa Jml iterasi dan Learning Rate yang bisa memberikan hasil/MSE terbaik?

JAWABAN 4:

1. Learning rate 0.1 dan Iterasi 500
2. Error Memorization (Train MSE): 726.99, Error Generalization (Test MSE): 757.57
3. Tabel Hasil Eksperimen

Eksperimen	Jml Iterasi	Learning Rate	MSE Train	MSE Test
1	500	0.01	6164.31	6403.56
2	1000	0.01	4371.23	4560.57
3	500	0.1	726.99	757.57
4	1000	0.1	360.73	371.51
5	500	0.5	315.11	321.12
6	1000	0.5	313.51	319.17

1. Jml Iterasi: 1000, Learning Rate: 0.5

##3. Eksperimen Efek Multiple Feature

###a. Training Model dengan Menggunakan Hanya Fitur 1 dan Evaluasi

Catatan: Gunakan Jml Iterasi dan Learning Rate terbaik Pada Hasil Sebelumnya

```
#Ambil fitur 1 saja
#[0,1] artinya kolom 0 dan 1, kolom 0 selalu kita pakai karena nilai
bias
idx_fitur = [0,1]
X_fitur_1 = X_train_sungguhan[:,idx_fitur]
X_val_fitur_1 =  X_val[:,idx_fitur]
#Train Model
w, hist_train, hist_val = train(X_fitur_1, y_train_sungguhan,
X_val_fitur_1, y_val, 0.1, 500)

#Evaluasi dengan data training
hasil_prediksi = predict(w,X_train_sungguhan[:,idx_fitur] )
error = mse(y_train_sungguhan, hasil_prediksi)
print("Error Trainig: ", error)

#Evaluasi dengan data TEsting
hasil_prediksi = predict(w,X_test[:,idx_fitur] )
error = mse(y_test, hasil_prediksi)
print("Error Testing: ", error)
```

```
Iterasi ke- 0   Train Loss:  115496.17792327816   Val Loss:
115408.38289412274
Iterasi ke- 1   Train Loss:  89941.63944769316   Val Loss:
89900.24735099453
```

```
Iterasi ke- 2   Train Loss:  70195.97107829261   Val Loss:
70186.34026965835
Iterasi ke- 3   Train Loss:  54938.452064962206   Val Loss:
54949.758598974375
Iterasi ke- 4   Train Loss:  43148.64576045114   Val Loss:
43172.958305436114
Iterasi ke- 5   Train Loss:  34038.12587580121   Val Loss:
34069.697607481234
Iterasi ke- 6   Train Loss:  26997.725716055855   Val Loss:
27032.45308438085
Iterasi ke- 7   Train Loss:  21556.781031948278   Val Loss:
21591.790777188282
Iterasi ke- 8   Train Loss:  17351.639573064316   Val Loss:
17384.97422812927
Iterasi ke- 9   Train Loss:  14101.33042971042   Val Loss:
14131.709382744786
Iterasi ke- 10   Train Loss:  11588.765286481137   Val Loss:
11615.403752252578
Iterasi ke- 11   Train Loss:  9646.213830876377   Val Loss:
9668.686147556315
Iterasi ke- 12   Train Loss:  8144.0815286922425   Val Loss:
8162.218333428796
Iterasi ke- 13   Train Loss:  6982.238927416709   Val Loss:
6996.0501820982945
Iterasi ke- 14   Train Loss:  6083.322362442179   Val Loss:
6092.940064320907
Iterasi ke- 15   Train Loss:  5387.557840393437   Val Loss:
5393.19368747964
Iterasi ke- 16   Train Loss:  4848.761784183241   Val Loss:
4850.6761703615175
Iterasi ke- 17   Train Loss:  4431.251064001833   Val Loss:
4429.73062917996
Iterasi ke- 18   Train Loss:  4107.455575504913   Val Loss:
4102.797190091521
Iterasi ke- 19   Train Loss:  3856.073631375891   Val Loss:
3848.573196991145
Iterasi ke- 20   Train Loss:  3660.6467501329084   Val Loss:
3650.591584416664
Iterasi ke- 21   Train Loss:  3508.458486415348   Val Loss:
3496.1223559059376
Iterasi ke- 22   Train Loss:  3389.6836274369175   Val Loss:
3375.3237194228577
Iterasi ke- 23   Train Loss:  3296.730831393955   Val Loss:
3280.5861293344146
Iterasi ke- 24   Train Loss:  3223.734726124468   Val Loss:
3206.0253859864088
Iterasi ke- 25   Train Loss:  3166.1634861623993   Val Loss:
3147.0909123095275
Iterasi ke- 26   Train Loss:  3120.5156325765015   Val Loss:
```

```
3100.263028970394
Iterasi ke- 27   Train Loss:  3084.085769559504   Val Loss:
3062.8190006214477
Iterasi ke- 28   Train Loss:  3054.7835840437683   Val Loss:
3032.6522239080236
Iterasi ke- 29   Train Loss:  3030.993998257559   Val Loss:
3008.1324806555695
Iterasi ke- 30   Train Loss:  3011.469118531598   Val Loss:
2987.997924745333
Iterasi ke- 31   Train Loss:  2995.244751038571   Val Loss:
2971.271592219423
Iterasi ke- 32   Train Loss:  2981.5758988335165   Val Loss:
2957.19686301367
Iterasi ke- 33   Train Loss:  2969.8869245337874   Val Loss:
2945.1875690201678
Iterasi ke- 34   Train Loss:  2959.7330442025514   Val Loss:
2934.7894216286777
Iterasi ke- 35   Train Loss:  2950.770576129963   Val Loss:
2925.650187928999
Iterasi ke- 36   Train Loss:  2942.733953964905   Val Loss:
2917.4966289380977
Iterasi ke- 37   Train Loss:  2935.417966228648   Val Loss:
2910.116664614094
Iterasi ke- 38   Train Loss:  2928.6640339201685   Val Loss:
2903.34557921971
Iterasi ke- 39   Train Loss:  2922.3496080971763   Val Loss:
2897.0553501236227
Iterasi ke- 40   Train Loss:  2916.3799780631684   Val Loss:
2891.1463914000656
Iterasi ke- 41   Train Loss:  2910.6819420759234   Val Loss:
2885.54116452879
Iterasi ke- 42   Train Loss:  2905.1989171074015   Val Loss:
2880.179232867611
Iterasi ke- 43   Train Loss:  2899.887160466822   Val Loss:
2875.0134326808643
Iterasi ke- 44   Train Loss:  2894.712850489436   Val Loss:
2870.006907782169
Iterasi ke- 45   Train Loss:  2889.64983097054   Val Loss:
2865.1308122517285
Iterasi ke- 46   Train Loss:  2884.6778684330557   Val Loss:
2860.3625300517656
Iterasi ke- 47   Train Loss:  2879.7813056288896   Val Loss:
2855.6842946515194
Iterasi ke- 48   Train Loss:  2874.9480211848327   Val Loss:
2851.082118275133
Iterasi ke- 49   Train Loss:  2870.1686257868046   Val Loss:
2846.5449608707268
Iterasi ke- 50   Train Loss:  2865.4358411221915   Val Loss:
2842.064084734083
```

```
Iterasi ke- 51   Train Loss:  2860.7440200276933   Val Loss:
2837.632552961931
Iterasi ke- 52   Train Loss:  2856.0887757376527   Val Loss:
2833.2448393740874
Iterasi ke- 53   Train Loss:  2851.4666954273703   Val Loss:
2828.896524861419
Iterasi ke- 54   Train Loss:  2846.8751188857786   Val Loss:
2824.584060775226
Iterasi ke- 55   Train Loss:  2842.31196750941   Val Loss:
2820.3045843497825
Iterasi ke- 56   Train Loss:  2837.775612176425   Val Loss:
2816.0557745346387
Iterasi ke- 57   Train Loss:  2833.264771160784   Val Loss:
2811.835739231731
Iterasi ke- 58   Train Loss:  2828.7784312565145   Val Loss:
2807.6429269583123
Iterasi ke- 59   Train Loss:  2824.3157868349517   Val Loss:
2803.4760575245728
Iterasi ke- 60   Train Loss:  2819.876192757636   Val Loss:
2799.3340675283716
Iterasi ke- 61   Train Loss:  2815.4591279946026   Val Loss:
2795.2160674091824
Iterasi ke- 62   Train Loss:  2811.064167514036   Val Loss:
2791.121307530989
Iterasi ke- 63   Train Loss:  2806.690960562696   Val Loss:
2787.0491513277284
Iterasi ke- 64   Train Loss:  2802.3392138840736   Val Loss:
2782.9990539817823
Iterasi ke- 65   Train Loss:  2798.0086787516334   Val Loss:
2778.9705454448035
Iterasi ke- 66   Train Loss:  2793.6991409497114   Val Loss:
2774.963216873018
Iterasi ke- 67   Train Loss:  2789.4104130319047   Val Loss:
2770.9767097530885
Iterasi ke- 68   Train Loss:  2785.142328339119   Val Loss:
2767.010707153075
Iterasi ke- 69   Train Loss:  2780.8947363771945   Val Loss:
2763.0649266561513
Iterasi ke- 70   Train Loss:  2776.667499245006   Val Loss:
2759.139114630508
Iterasi ke- 71   Train Loss:  2772.460488874186   Val Loss:
2755.233041563431
Iterasi ke- 72   Train Loss:  2768.273584895946   Val Loss:
2751.346498245661
Iterasi ke- 73   Train Loss:  2764.1066729924196   Val Loss:
2747.479292637457
Iterasi ke- 74   Train Loss:  2759.9596436223655   Val Loss:
2743.631247283221
Iterasi ke- 75   Train Loss:  2755.8323910361137   Val Loss:
```

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2739.8021971692156
Iterasi ke- 76   Train Loss:  2751.7248125139968   Val Loss:
2735.991987940657
Iterasi ke- 77   Train Loss:  2747.636807777462   Val Loss:
2732.2004744114606
Iterasi ke- 78   Train Loss:  2743.5682785335907   Val Loss:
2728.4275193133412
Iterasi ke- 79   Train Loss:  2739.5191281227094   Val Loss:
2724.67299224153
Iterasi ke- 80   Train Loss:  2735.489261245649   Val Loss:
2720.936768762704
Iterasi ke- 81   Train Loss:  2731.4785837525433   Val Loss:
2717.2187296573334
Iterasi ke- 82   Train Loss:  2727.487002479184   Val Loss:
2713.51876027391
Iterasi ke- 83   Train Loss:  2723.5144251201136   Val Loss:
2709.8367499766928
Iterasi ke- 84   Train Loss:  2719.560760130112   Val Loss:
2706.1725916719424
Iterasi ke- 85   Train Loss:  2715.625916647626   Val Loss:
2702.526181400309
Iterasi ke- 86   Train Loss:  2711.7098044351364   Val Loss:
2698.8974179851957
Iterasi ke- 87   Train Loss:  2707.8123338326413   Val Loss:
2695.286202728644
Iterasi ke- 88   Train Loss:  2703.933415721256   Val Loss:
2691.6924391477155
Iterasi ke- 89   Train Loss:  2700.072961494637   Val Loss:
2688.116032745514
Iterasi ke- 90   Train Loss:  2696.2308830364523   Val Loss:
2684.556890811888
Iterasi ke- 91   Train Loss:  2692.40709270253   Val Loss:
2681.014922249676
Iterasi ke- 92   Train Loss:  2688.6015033066187   Val Loss:
2677.490037422974
Iterasi ke- 93   Train Loss:  2684.8140281089436   Val Loss:
2673.982148024427
Iterasi ke- 94   Train Loss:  2681.04458080692   Val Loss:
2670.491166959009
Iterasi ke- 95   Train Loss:  2677.2930755275406   Val Loss:
2667.0170082421128
Iterasi ke- 96   Train Loss:  2673.559426821062   Val Loss:
2663.559586910071
Iterasi ke- 97   Train Loss:  2669.8435496556835   Val Loss:
2660.118818941517
Iterasi ke- 98   Train Loss:  2666.1453594130076   Val Loss:
2656.6946211881937
Iterasi ke- 99   Train Loss:  2662.4647718841024   Val Loss:
2653.2869113140046
```

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Iterasi ke- 100   Train Loss:  2658.801703266029   Val Loss:
2649.8956077412904
Iterasi ke- 101   Train Loss:  2655.1560701587314   Val Loss:
2646.520629603409
Iterasi ke- 102   Train Loss:  2651.527789562212   Val Loss:
2643.161896702847
Iterasi ke- 103   Train Loss:  2647.916778873922   Val Loss:
2639.8193294741864
Iterasi ke- 104   Train Loss:  2644.32295588633   Val Loss:
2636.492848951314
Iterasi ke- 105   Train Loss:  2640.746238784618   Val Loss:
2633.1823767383808
Iterasi ke- 106   Train Loss:  2637.1865461444945   Val Loss:
2629.887834984024
Iterasi ke- 107   Train Loss:  2633.643796930079   Val Loss:
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Iterasi ke- 108   Train Loss:  2630.1179104918633   Val Loss:
2623.34623403328
Iterasi ke- 109   Train Loss:  2626.608806564725   Val Loss:
2620.0990216630394
Iterasi ke- 110   Train Loss:  2623.116405265975   Val Loss:
2616.8674333693953
Iterasi ke- 111   Train Loss:  2619.6406270934576   Val Loss:
2613.6513937265336
Iterasi ke- 112   Train Loss:  2616.181392923662   Val Loss:
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Iterasi ke- 113   Train Loss:  2612.7386240098713   Val Loss:
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Iterasi ke- 114   Train Loss:  2609.3122419803276   Val Loss:
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Iterasi ke- 115   Train Loss:  2605.902168836415   Val Loss:
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Iterasi ke- 116   Train Loss:  2602.508326950857   Val Loss:
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Iterasi ke- 117   Train Loss:  2599.1306390659324   Val Loss:
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Iterasi ke- 118   Train Loss:  2595.769028291702   Val Loss:
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Iterasi ke- 119   Train Loss:  2592.423418104244   Val Loss:
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Iterasi ke- 120   Train Loss:  2589.093732343906   Val Loss:
2585.394490425985
Iterasi ke- 121   Train Loss:  2585.7798952135618   Val Loss:
2582.32988068987
Iterasi ke- 122   Train Loss:  2582.4818312768843   Val Loss:
2579.2800168817203
Iterasi ke- 123   Train Loss:  2579.1994654566183   Val Loss:
2576.2448282905784
Iterasi ke- 124   Train Loss:  2575.9327230328754   Val Loss:
```

```
2573.224244560455
Iterasi ke- 125   Train Loss:  2572.681529641423   Val Loss:
2570.2181956865597
Iterasi ke- 126   Train Loss:  2569.4458112719976   Val Loss:
2567.226612011795
Iterasi ke- 127   Train Loss:  2566.225494266612   Val Loss:
2564.24942422349
Iterasi ke- 128   Train Loss:  2563.0205053178797   Val Loss:
2561.286563350332
Iterasi ke- 129   Train Loss:  2559.830771467347   Val Loss:
2558.3379607594898
Iterasi ke- 130   Train Loss:  2556.6562201038273   Val Loss:
2555.4035481538936
Iterasi ke- 131   Train Loss:  2553.496778961754   Val Loss:
2552.483257569661
Iterasi ke- 132   Train Loss:  2550.352376119529   Val Loss:
2549.577021373656
Iterasi ke- 133   Train Loss:  2547.222939997887   Val Loss:
2546.6847722611506
Iterasi ke- 134   Train Loss:  2544.1083993582665   Val Loss:
2543.8064432535934
Iterasi ke- 135   Train Loss:  2541.00868330119   Val Loss:
2540.9419676964726
Iterasi ke- 136   Train Loss:  2537.9237212646453   Val Loss:
2538.0912792572462
Iterasi ke- 137   Train Loss:  2534.8534430224818   Val Loss:
2535.2543119233596
Iterasi ke- 138   Train Loss:  2531.797778682812   Val Loss:
2532.4310000003125
Iterasi ke- 139   Train Loss:  2528.756658686419   Val Loss:
2529.6212781097906
Iterasi ke- 140   Train Loss:  2525.730013805175   Val Loss:
2526.825081187845
Iterasi ke- 141   Train Loss:  2522.7177751404643   Val Loss:
2524.042344483122
Iterasi ke- 142   Train Loss:  2519.719874121613   Val Loss:
2521.2730035551253
Iterasi ke- 143   Train Loss:  2516.7362425043316   Val Loss:
2518.5169942725256
Iterasi ke- 144   Train Loss:  2513.7668123691587   Val Loss:
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Iterasi ke- 145   Train Loss:  2510.8115161199157   Val Loss:
2513.04471565408
Iterasi ke- 146   Train Loss:  2507.8702864821685   Val Loss:
2510.3283195865893
Iterasi ke- 147   Train Loss:  2504.943056501697   Val Loss:
2507.6250016980234
Iterasi ke- 148   Train Loss:  2502.029759542967   Val Loss:
2504.9346993785125
```

```
Iterasi ke- 149   Train Loss:  2499.130329287618   Val Loss:
2502.2573503177837
Iterasi ke- 150   Train Loss:  2496.244699732951   Val Loss:
2499.5928925036433
Iterasi ke- 151   Train Loss:  2493.372805190426   Val Loss:
2496.9412642204793
Iterasi ke- 152   Train Loss:  2490.5145802841666   Val Loss:
2494.30240404778
Iterasi ke- 153   Train Loss:  2487.669959949475   Val Loss:
2491.6762508586626
Iterasi ke- 154   Train Loss:  2484.8388794313446   Val Loss:
2489.0627438184265
Iterasi ke- 155   Train Loss:  2482.0212742829926   Val Loss:
2486.4618223831117
Iterasi ke- 156   Train Loss:  2479.2170803643885   Val Loss:
2483.8734262980674
Iterasi ke- 157   Train Loss:  2476.426233840794   Val Loss:
2481.2974955965433
Iterasi ke- 158   Train Loss:  2473.648671181314   Val Loss:
2478.733970598281
Iterasi ke- 159   Train Loss:  2470.884329157446   Val Loss:
2476.18279190812
Iterasi ke- 160   Train Loss:  2468.1331448416427   Val Loss:
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Iterasi ke- 161   Train Loss:  2465.3950556058776   Val Loss:
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Iterasi ke- 162   Train Loss:  2462.669999120224   Val Loss:
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Iterasi ke- 163   Train Loss:  2459.957913351431   Val Loss:
2466.100362226537
Iterasi ke- 164   Train Loss:  2457.258736561513   Val Loss:
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Iterasi ke- 165   Train Loss:  2454.5724073063475   Val Loss:
2461.131701684063
Iterasi ke- 166   Train Loss:  2451.898864434269   Val Loss:
2458.665307746184
Iterasi ke- 167   Train Loss:  2449.2380470846856   Val Loss:
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Iterasi ke- 168   Train Loss:  2446.5898946866887   Val Loss:
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Iterasi ke- 169   Train Loss:  2443.954346957673   Val Loss:
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Iterasi ke- 170   Train Loss:  2441.3313439019676   Val Loss:
2448.9179769093457
Iterasi ke- 171   Train Loss:  2438.720825809469   Val Loss:
2446.5104230079864
Iterasi ke- 172   Train Loss:  2436.1227332542812   Val Loss:
2444.114468604044
Iterasi ke- 173   Train Loss:  2433.5370070933636   Val Loss:
```

```
2441.730058152351
Iterasi ke- 174   Train Loss:  2430.9635884651857   Val Loss:
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Iterasi ke- 175   Train Loss:  2428.402418788382   Val Loss:
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Iterasi ke- 176   Train Loss:  2425.8534397604285   Val Loss:
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Iterasi ke- 177   Train Loss:  2423.3165933563046   Val Loss:
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Iterasi ke- 178   Train Loss:  2420.791821827178   Val Loss:
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Iterasi ke- 179   Train Loss:  2418.2790676990903   Val Loss:
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Iterasi ke- 180   Train Loss:  2415.778273771647   Val Loss:
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Iterasi ke- 181   Train Loss:  2413.2893831167153   Val Loss:
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Iterasi ke- 182   Train Loss:  2410.8123390771298   Val Loss:
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Iterasi ke- 183   Train Loss:  2408.347085265401   Val Loss:
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Iterasi ke- 184   Train Loss:  2405.8935655624327   Val Loss:
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Iterasi ke- 185   Train Loss:  2403.451724116245   Val Loss:
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Iterasi ke- 186   Train Loss:  2401.021505340699   Val Loss:
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Iterasi ke- 187   Train Loss:  2398.60285391424   Val Loss:
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Iterasi ke- 188   Train Loss:  2396.1957147786256   Val Loss:
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Iterasi ke- 189   Train Loss:  2393.8000331376875   Val Loss:
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Iterasi ke- 190   Train Loss:  2391.415754456069   Val Loss:
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Iterasi ke- 191   Train Loss:  2389.0428244579953   Val Loss:
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Iterasi ke- 192   Train Loss:  2386.6811891260304   Val Loss:
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Iterasi ke- 193   Train Loss:  2384.330794699853   Val Loss:
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Iterasi ke- 194   Train Loss:  2381.991587675026   Val Loss:
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Iterasi ke- 195   Train Loss:  2379.6635148017886   Val Loss:
2392.084513371342
Iterasi ke- 196   Train Loss:  2377.346523083833   Val Loss:
2389.950877630923
Iterasi ke- 197   Train Loss:  2375.040559777104   Val Loss:
2387.8275294839036
```

```
Iterasi ke- 198   Train Loss:  2372.7455723885996   Val Loss:
2385.7144196490794
Iterasi ke- 199   Train Loss:  2370.4615086751714   Val Loss:
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Iterasi ke- 200   Train Loss:  2368.188316642339   Val Loss:
2381.5187189666917
Iterasi ke- 201   Train Loss:  2365.925944543106   Val Loss:
2379.436030728837
Iterasi ke- 202   Train Loss:  2363.6743408767793   Val Loss:
2377.3633860203713
Iterasi ke- 203   Train Loss:  2361.4334543878   Val Loss:
2375.3007367255195
Iterasi ke- 204   Train Loss:  2359.203234064574   Val Loss:
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Iterasi ke- 205   Train Loss:  2356.9836291383135   Val Loss:
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Iterasi ke- 206   Train Loss:  2354.77458908188   Val Loss:
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Iterasi ke- 207   Train Loss:  2352.576063608633   Val Loss:
2367.149139385499
Iterasi ke- 208   Train Loss:  2350.3880026712854   Val Loss:
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Iterasi ke- 209   Train Loss:  2348.210356460767   Val Loss:
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Iterasi ke- 210   Train Loss:  2346.0430754050876   Val Loss:
2361.138069492886
Iterasi ke- 211   Train Loss:  2343.8861101682105   Val Loss:
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Iterasi ke- 212   Train Loss:  2341.7394116489263   Val Loss:
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Iterasi ke- 213   Train Loss:  2339.60293097974   Val Loss:
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Iterasi ke- 214   Train Loss:  2337.4766195257553   Val Loss:
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Iterasi ke- 215   Train Loss:  2335.3604288835672   Val Loss:
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Iterasi ke- 216   Train Loss:  2333.254310880162   Val Loss:
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Iterasi ke- 217   Train Loss:  2331.1582175718186   Val Loss:
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Iterasi ke- 218   Train Loss:  2329.072101243021   Val Loss:
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Iterasi ke- 219   Train Loss:  2326.9959144053655   Val Loss:
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Iterasi ke- 220   Train Loss:  2324.9296097964875   Val Loss:
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Iterasi ke- 221   Train Loss:  2322.8731403789784   Val Loss:
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Iterasi ke- 222   Train Loss:  2320.82645933932   Val Loss:
```

```
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Iterasi ke- 223   Train Loss:  2318.7895200868174   Val Loss:
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Iterasi ke- 224   Train Loss:  2316.7622762525366   Val Loss:
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Iterasi ke- 225   Train Loss:  2314.744681688251   Val Loss:
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Iterasi ke- 226   Train Loss:  2312.7366904653895   Val Loss:
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Iterasi ke- 227   Train Loss:  2310.738256873992   Val Loss:
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Iterasi ke- 228   Train Loss:  2308.7493354216704   Val Loss:
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Iterasi ke- 229   Train Loss:  2306.7698808325677   Val Loss:
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Iterasi ke- 230   Train Loss:  2304.7998480463343   Val Loss:
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Iterasi ke- 231   Train Loss:  2302.8391922170954   Val Loss:
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Iterasi ke- 232   Train Loss:  2300.887868712437   Val Loss:
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Iterasi ke- 233   Train Loss:  2298.945833112383   Val Loss:
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Iterasi ke- 234   Train Loss:  2297.0130412083895   Val Loss:
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Iterasi ke- 235   Train Loss:  2295.0894490023366   Val Loss:
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Iterasi ke- 236   Train Loss:  2293.1750127055248   Val Loss:
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Iterasi ke- 237   Train Loss:  2291.2696887376796   Val Loss:
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Iterasi ke- 238   Train Loss:  2289.3734337259634   Val Loss:
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Iterasi ke- 239   Train Loss:  2287.4862045039804   Val Loss:
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Iterasi ke- 241   Train Loss:  2283.7386517899786   Val Loss:
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Iterasi ke- 242   Train Loss:  2281.8782429885805   Val Loss:
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Iterasi ke- 243   Train Loss:  2280.026689356216   Val Loss:
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Iterasi ke- 244   Train Loss:  2278.183948744075   Val Loss:
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Iterasi ke- 245   Train Loss:  2276.3499792039665   Val Loss:
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Iterasi ke- 246   Train Loss:  2274.5247389873634   Val Loss:
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Iterasi ke- 247   Train Loss:  2272.708186544455   Val Loss:
```

```
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Iterasi ke- 248   Train Loss:  2270.9002805231985   Val Loss:
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Iterasi ke- 249   Train Loss:  2269.1009797683773   Val Loss:
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Iterasi ke- 250   Train Loss:  2267.3102433206677   Val Loss:
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Iterasi ke- 251   Train Loss:  2265.528030415701   Val Loss:
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Iterasi ke- 252   Train Loss:  2263.7543004831427   Val Loss:
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Iterasi ke- 253   Train Loss:  2261.9890131457596   Val Loss:
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Iterasi ke- 254   Train Loss:  2260.2321282185108   Val Loss:
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Iterasi ke- 255   Train Loss:  2258.483605707626   Val Loss:
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Iterasi ke- 256   Train Loss:  2256.743405809698   Val Loss:
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Iterasi ke- 257   Train Loss:  2255.0114889107726   Val Loss:
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Iterasi ke- 258   Train Loss:  2253.287815585454   Val Loss:
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Iterasi ke- 259   Train Loss:  2251.5723465960027   Val Loss:
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Iterasi ke- 260   Train Loss:  2249.86504289144   Val Loss:
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Iterasi ke- 261   Train Loss:  2248.1658656066643   Val Loss:
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Iterasi ke- 262   Train Loss:  2246.4747760615646   Val Loss:
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Iterasi ke- 263   Train Loss:  2244.7917357601373   Val Loss:
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Iterasi ke- 264   Train Loss:  2243.1167063896137   Val Loss:
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Iterasi ke- 265   Train Loss:  2241.4496498195854   Val Loss:
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Iterasi ke- 266   Train Loss:  2239.7905281011363   Val Loss:
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Iterasi ke- 267   Train Loss:  2238.139303465981   Val Loss:
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Iterasi ke- 268   Train Loss:  2236.4959383256023   Val Loss:
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Iterasi ke- 269   Train Loss:  2234.860395270399   Val Loss:
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Iterasi ke- 270   Train Loss:  2233.232637068826   Val Loss:
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Iterasi ke- 271   Train Loss:  2231.6126266665615   Val Loss:
2256.0656657602494
```

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Iterasi ke- 272   Train Loss:  2230.000327185648   Val Loss:
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Iterasi ke- 273   Train Loss:  2228.395701923662   Val Loss:
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Iterasi ke- 274   Train Loss:  2226.7987143528753   Val Loss:
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Iterasi ke- 275   Train Loss:  2225.209328119427   Val Loss:
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Iterasi ke- 276   Train Loss:  2223.62750704249   Val Loss:
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Iterasi ke- 277   Train Loss:  2222.053215113451   Val Loss:
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Iterasi ke- 278   Train Loss:  2220.486416495092   Val Loss:
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Iterasi ke- 279   Train Loss:  2218.9270755207717   Val Loss:
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Iterasi ke- 280   Train Loss:  2217.3751566936157   Val Loss:
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Iterasi ke- 281   Train Loss:  2215.8306246857073   Val Loss:
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Iterasi ke- 282   Train Loss:  2214.2934443372815   Val Loss:
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Iterasi ke- 283   Train Loss:  2212.7635806559306   Val Loss:
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Iterasi ke- 284   Train Loss:  2211.2409988157992   Val Loss:
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Iterasi ke- 285   Train Loss:  2209.7256641568006   Val Loss:
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Iterasi ke- 286   Train Loss:  2208.217542183819   Val Loss:
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Iterasi ke- 287   Train Loss:  2206.7165985659303   Val Loss:
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Iterasi ke- 288   Train Loss:  2205.2227991356185   Val Loss:
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Iterasi ke- 289   Train Loss:  2203.736109887998   Val Loss:
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Iterasi ke- 290   Train Loss:  2202.256496980039   Val Loss:
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Iterasi ke- 291   Train Loss:  2200.7839267297995   Val Loss:
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Iterasi ke- 292   Train Loss:  2199.318365615654   Val Loss:
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Iterasi ke- 293   Train Loss:  2197.8597802755344   Val Loss:
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Iterasi ke- 294   Train Loss:  2196.408137506171   Val Loss:
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Iterasi ke- 295   Train Loss:  2194.9634042623316   Val Loss:
2222.520013914701
Iterasi ke- 296   Train Loss:  2193.5255476560756   Val Loss:
```

```
2221.2052212712497
Iterasi ke- 297   Train Loss:  2192.0945349559997   Val Loss:
2219.8967906956514
Iterasi ke- 298   Train Loss:  2190.670333586498   Val Loss:
2218.5946916581356
Iterasi ke- 299   Train Loss:  2189.2529111270173   Val Loss:
2217.298893774837
Iterasi ke- 300   Train Loss:  2187.842235311318   Val Loss:
2216.009366807098
Iterasi ke- 301   Train Loss:  2186.4382740267415   Val Loss:
2214.726080660782
Iterasi ke- 302   Train Loss:  2185.0409953134817   Val Loss:
2213.4490053855766
Iterasi ke- 303   Train Loss:  2183.650367363852   Val Loss:
2212.1781111743126
Iterasi ke- 304   Train Loss:  2182.266358521565   Val Loss:
2210.913368362279
Iterasi ke- 305   Train Loss:  2180.88893728101   Val Loss:
2209.6547474265467
Iterasi ke- 306   Train Loss:  2179.518072286538   Val Loss:
2208.402218985288
Iterasi ke- 307   Train Loss:  2178.1537323317452   Val Loss:
2207.155753797106
Iterasi ke- 308   Train Loss:  2176.795886358764   Val Loss:
2205.9153227603647
Iterasi ke- 309   Train Loss:  2175.444503457558   Val Loss:
2204.680896912521
Iterasi ke- 310   Train Loss:  2174.0995528652134   Val Loss:
2203.452447429465
Iterasi ke- 311   Train Loss:  2172.761003965243   Val Loss:
2202.2299456248534
Iterasi ke- 312   Train Loss:  2171.428826286888   Val Loss:
2201.01336294946
Iterasi ke- 313   Train Loss:  2170.1029895044258   Val Loss:
2199.802670990515
Iterasi ke- 314   Train Loss:  2168.783463436474   Val Loss:
2198.5978414710567
Iterasi ke- 315   Train Loss:  2167.470218045311   Val Loss:
2197.3988462492857
Iterasi ke- 316   Train Loss:  2166.1632234361878   Val Loss:
2196.2056573179184
Iterasi ke- 317   Train Loss:  2164.862449856648   Val Loss:
2195.018246803543
Iterasi ke- 318   Train Loss:  2163.56786769585   Val Loss:
2193.836586965987
Iterasi ke- 319   Train Loss:  2162.279447483896   Val Loss:
2192.6606501976744
Iterasi ke- 320   Train Loss:  2160.997159891157   Val Loss:
2191.490409022999
```

```
Iterasi ke- 321   Train Loss:  2159.720975727608   Val Loss:
2190.325836097692
Iterasi ke- 322   Train Loss:  2158.450865942163   Val Loss:
2189.166904208197
Iterasi ke- 323   Train Loss:  2157.186801622012   Val Loss:
2188.013586271045
Iterasi ke- 324   Train Loss:  2155.9287539919646   Val Loss:
2186.8658553322352
Iterasi ke- 325   Train Loss:  2154.6766944137953   Val Loss:
2185.7236845666152
Iterasi ke- 326   Train Loss:  2153.43059438559   Val Loss:
2184.5870472772726
Iterasi ke- 327   Train Loss:  2152.190425541098   Val Loss:
2183.455916894912
Iterasi ke- 328   Train Loss:  2150.9561596490867   Val Loss:
2182.3302669772606
Iterasi ke- 329   Train Loss:  2149.727768612697   Val Loss:
2181.210071208448
Iterasi ke- 330   Train Loss:  2148.5052244688077   Val Loss:
2180.095303398414
Iterasi ke- 331   Train Loss:  2147.288499387394   Val Loss:
2178.9859374823036
Iterasi ke- 332   Train Loss:  2146.0775656708965   Val Loss:
2177.8819475198698
Iterasi ke- 333   Train Loss:  2144.872395753593   Val Loss:
2176.7833076948796
Iterasi ke- 334   Train Loss:  2143.6729622009666   Val Loss:
2175.689992314525
Iterasi ke- 335   Train Loss:  2142.479237709082   Val Loss:
2174.60197580883
Iterasi ke- 336   Train Loss:  2141.2911951039678   Val Loss:
2173.519232730067
Iterasi ke- 337   Train Loss:  2140.1088073409937   Val Loss:
2172.4417377521745
Iterasi ke- 338   Train Loss:  2138.9320475042564   Val Loss:
2171.369465670173
Iterasi ke- 339   Train Loss:  2137.760888805967   Val Loss:
2170.302391399593
Iterasi ke- 340   Train Loss:  2136.595304585842   Val Loss:
2169.240489975895
Iterasi ke- 341   Train Loss:  2135.4352683104935   Val Loss:
2168.1837365538995
Iterasi ke- 342   Train Loss:  2134.2807535728284   Val Loss:
2167.132106407219
Iterasi ke- 343   Train Loss:  2133.131734091446   Val Loss:
2166.085574927687
Iterasi ke- 344   Train Loss:  2131.9881837100397   Val Loss:
2165.0441176248
Iterasi ke- 345   Train Loss:  2130.850076396802   Val Loss:
```

```
2164.0077101251513
Iterasi ke- 346   Train Loss:  2129.71738624383   Val Loss:
2162.976328171875
Iterasi ke- 347   Train Loss:  2128.590087466538   Val Loss:
2161.949947624087
Iterasi ke- 348   Train Loss:  2127.4681544030705   Val Loss:
2160.9285444563366
Iterasi ke- 349   Train Loss:  2126.3515615137167   Val Loss:
2159.9120947580514
Iterasi ke- 350   Train Loss:  2125.2402833803294   Val Loss:
2158.900574732991
Iterasi ke- 351   Train Loss:  2124.134294705747   Val Loss:
2157.893960698703
Iterasi ke- 352   Train Loss:  2123.0335703132164   Val Loss:
2156.8922290859755
Iterasi ke- 353   Train Loss:  2121.9380851458222   Val Loss:
2155.895356438303
Iterasi ke- 354   Train Loss:  2120.847814265914   Val Loss:
2154.9033194113454
Iterasi ke- 355   Train Loss:  2119.762732854539   Val Loss:
2153.916094772391
Iterasi ke- 356   Train Loss:  2118.682816210879   Val Loss:
2152.9336593998273
Iterasi ke- 357   Train Loss:  2117.6080397516853   Val Loss:
2151.95599028261
Iterasi ke- 358   Train Loss:  2116.538379010721   Val Loss:
2150.9830645197358
Iterasi ke- 359   Train Loss:  2115.4738096382043   Val Loss:
2150.014859319716
Iterasi ke- 360   Train Loss:  2114.414307400252   Val Loss:
2149.051352000054
Iterasi ke- 361   Train Loss:  2113.3598481783297   Val Loss:
2148.092519986728
Iterasi ke- 362   Train Loss:  2112.310407968704   Val Loss:
2147.1383408136703
Iterasi ke- 363   Train Loss:  2111.265962881893   Val Loss:
2146.1887921222556
Iterasi ke- 364   Train Loss:  2110.226489142122   Val Loss:
2145.243851660783
Iterasi ke- 365   Train Loss:  2109.191963086788   Val Loss:
2144.303497283972
Iterasi ke- 366   Train Loss:  2108.1623611659147   Val Loss:
2143.3677069524515
Iterasi ke- 367   Train Loss:  2107.1376599416194   Val Loss:
2142.4364587322543
Iterasi ke- 368   Train Loss:  2106.1178360875797   Val Loss:
2141.509730794315
Iterasi ke- 369   Train Loss:  2105.1028663885004   Val Loss:
2140.58750141397
```

```
Iterasi ke- 370   Train Loss:  2104.092727739588   Val Loss:
2139.669748970459
Iterasi ke- 371   Train Loss:  2103.087397146022   Val Loss:
2138.7564519464277
Iterasi ke- 372   Train Loss:  2102.086851722432   Val Loss:
2137.847588927436
Iterasi ke- 373   Train Loss:  2101.0910686923803   Val Loss:
2136.9431386014685
Iterasi ke- 374   Train Loss:  2100.100025387836   Val Loss:
2136.043079758442
Iterasi ke- 375   Train Loss:  2099.1136992486677   Val Loss:
2135.1473912897222
Iterasi ke- 376   Train Loss:  2098.1320678221223   Val Loss:
2134.2560521876385
Iterasi ke- 377   Train Loss:  2097.15510876232   Val Loss:
2133.3690415450014
Iterasi ke- 378   Train Loss:  2096.182799829741   Val Loss:
2132.4863385546264
Iterasi ke- 379   Train Loss:  2095.2151188907214   Val Loss:
2131.607922508852
Iterasi ke- 380   Train Loss:  2094.25204391695   Val Loss:
2130.7337727990684
Iterasi ke- 381   Train Loss:  2093.293552984964   Val Loss:
2129.8638689152435
Iterasi ke- 382   Train Loss:  2092.339624275655   Val Loss:
2128.998190445453
Iterasi ke- 383   Train Loss:  2091.390236073765   Val Loss:
2128.1367170754133
Iterasi ke- 384   Train Loss:  2090.445366767401   Val Loss:
2127.279428588011
Iterasi ke- 385   Train Loss:  2089.5049948475335   Val Loss:
2126.426304862847
Iterasi ke- 386   Train Loss:  2088.569098907515   Val Loss:
2125.577325875767
Iterasi ke- 387   Train Loss:  2087.6376576425873   Val Loss:
2124.732471698408
Iterasi ke- 388   Train Loss:  2086.7106498494   Val Loss:
2123.891722497738
Iterasi ke- 389   Train Loss:  2085.7880544255254   Val Loss:
2123.0550585356036
Iterasi ke- 390   Train Loss:  2084.8698503689798   Val Loss:
2122.222460168274
Iterasi ke- 391   Train Loss:  2083.9560167777436   Val Loss:
2121.393907845994
Iterasi ke- 392   Train Loss:  2083.0465328492874   Val Loss:
2120.569382112533
Iterasi ke- 393   Train Loss:  2082.141377880098   Val Loss:
2119.7488636047397
Iterasi ke- 394   Train Loss:  2081.2405312652054   Val Loss:
```

```
2118.9323330520974
Iterasi ke- 395   Train Loss:  2080.3439724977165   Val Loss:
2118.119771276284
Iterasi ke- 396   Train Loss:  2079.4516811683466   Val Loss:
2117.311159190729
Iterasi ke- 397   Train Loss:  2078.5636369649537   Val Loss:
2116.5064778001774
Iterasi ke- 398   Train Loss:  2077.6798196720792   Val Loss:
2115.7057082002534
Iterasi ke- 399   Train Loss:  2076.800209170485   Val Loss:
2114.908831577028
Iterasi ke- 400   Train Loss:  2075.9247854366963   Val Loss:
2114.115829206585
Iterasi ke- 401   Train Loss:  2075.053528542547   Val Loss:
2113.3266824545944
Iterasi ke- 402   Train Loss:  2074.186418654725   Val Loss:
2112.541372775882
Iterasi ke- 403   Train Loss:  2073.3234360343195   Val Loss:
2111.759881714005
Iterasi ke- 404   Train Loss:  2072.4645610363737   Val Loss:
2110.9821909008306
Iterasi ke- 405   Train Loss:  2071.609774109437   Val Loss:
2110.2082820561113
Iterasi ke- 406   Train Loss:  2070.7590557951194   Val Loss:
2109.4381369870675
Iterasi ke- 407   Train Loss:  2069.9123867276485   Val Loss:
2108.6717375879684
Iterasi ke- 408   Train Loss:  2069.069747633431   Val Loss:
2107.9090658397204
Iterasi ke- 409   Train Loss:  2068.231119330609   Val Loss:
2107.1501038094475
Iterasi ke- 410   Train Loss:  2067.3964827286286   Val Loss:
2106.3948336500857
Iterasi ke- 411   Train Loss:  2066.5658188278026   Val Loss:
2105.6432375999716
Iterasi ke- 412   Train Loss:  2065.739108718878   Val Loss:
2104.895297982431
Iterasi ke- 413   Train Loss:  2064.916333582606   Val Loss:
2104.150997205381
Iterasi ke- 414   Train Loss:  2064.097474689314   Val Loss:
2103.4103177609177
Iterasi ke- 415   Train Loss:  2063.2825133984793   Val Loss:
2102.6732422249215
Iterasi ke- 416   Train Loss:  2062.4714311583034   Val Loss:
2101.939753256653
Iterasi ke- 417   Train Loss:  2061.6642095052903   Val Loss:
2101.2098335983583
Iterasi ke- 418   Train Loss:  2060.8608300638284   Val Loss:
2100.4834660748684
```

```
Iterasi ke- 419   Train Loss:  2060.0612745457674   Val Loss:
2099.7606335932096
Iterasi ke- 420   Train Loss:  2059.265524750008   Val Loss:
2099.041319142209
Iterasi ke- 421   Train Loss:  2058.473562562082   Val Loss:
2098.3255057921037
Iterasi ke- 422   Train Loss:  2057.685369953744   Val Loss:
2097.6131766941553
Iterasi ke- 423   Train Loss:  2056.9009289825585   Val Loss:
2096.904315080257
Iterasi ke- 424   Train Loss:  2056.120221791493   Val Loss:
2096.1989042625564
Iterasi ke- 425   Train Loss:  2055.3432306085097   Val Loss:
2095.4969276330667
Iterasi ke- 426   Train Loss:  2054.5699377461633   Val Loss:
2094.7983686632883
Iterasi ke- 427   Train Loss:  2053.8003256011966   Val Loss:
2094.1032109038283
Iterasi ke- 428   Train Loss:  2053.0343766541405   Val Loss:
2093.411437984024
Iterasi ke- 429   Train Loss:  2052.2720734689156   Val Loss:
2092.723033611565
Iterasi ke- 430   Train Loss:  2051.513398692433   Val Loss:
2092.0379815721226
Iterasi ke- 431   Train Loss:  2050.7583350542027   Val Loss:
2091.356265728973
Iterasi ke- 432   Train Loss:  2050.006865365938   Val Loss:
2090.6778700226328
Iterasi ke- 433   Train Loss:  2049.258972521166   Val Loss:
2090.002778470484
Iterasi ke- 434   Train Loss:  2048.5146394948347   Val Loss:
2089.330975166413
Iterasi ke- 435   Train Loss:  2047.7738493429304   Val Loss:
2088.6624442804396
Iterasi ke- 436   Train Loss:  2047.036585202088   Val Loss:
2087.997170058359
Iterasi ke- 437   Train Loss:  2046.3028302892096   Val Loss:
2087.3351368213766
Iterasi ke- 438   Train Loss:  2045.5725679010807   Val Loss:
2086.6763289657506
Iterasi ke- 439   Train Loss:  2044.845781413991   Val Loss:
2086.0207309624298
Iterasi ke- 440   Train Loss:  2044.1224542833554   Val Loss:
2085.3683273567017
Iterasi ke- 441   Train Loss:  2043.4025700433383   Val Loss:
2084.719102767834
Iterasi ke- 442   Train Loss:  2042.6861123064775   Val Loss:
2084.073041888726
Iterasi ke- 443   Train Loss:  2041.9730647633125   Val Loss:
```

```
2083.43012948555
Iterasi ke- 444   Train Loss:  2041.2634111820116   Val Loss:
2082.790350397409
Iterasi ke- 445   Train Loss:  2040.5571354080041   Val Loss:
2082.1536895359827
Iterasi ke- 446   Train Loss:  2039.8542213636115   Val Loss:
2081.5201318851878
Iterasi ke- 447   Train Loss:  2039.154653047682   Val Loss:
2080.889662500823
Iterasi ke- 448   Train Loss:  2038.458414535225   Val Loss:
2080.262266510238
Iterasi ke- 449   Train Loss:  2037.7654899770519   Val Loss:
2079.6379291119806
Iterasi ke- 450   Train Loss:  2037.0758635994105   Val Loss:
2079.016635575466
Iterasi ke- 451   Train Loss:  2036.3895197036297   Val Loss:
2078.3983712406325
Iterasi ke- 452   Train Loss:  2035.7064426657614   Val Loss:
2077.7831215176075
Iterasi ke- 453   Train Loss:  2035.0266169362233   Val Loss:
2077.170871886371
Iterasi ke- 454   Train Loss:  2034.350027039446   Val Loss:
2076.561607896425
Iterasi ke- 455   Train Loss:  2033.6766575735214   Val Loss:
2075.9553151664572
Iterasi ke- 456   Train Loss:  2033.0064932098514   Val Loss:
2075.3519793840146
Iterasi ke- 457   Train Loss:  2032.339518692798   Val Loss:
2074.751586305173
Iterasi ke- 458   Train Loss:  2031.6757188393376   Val Loss:
2074.1541217542103
Iterasi ke- 459   Train Loss:  2031.0150785387143   Val Loss:
2073.5595716232806
Iterasi ke- 460   Train Loss:  2030.3575827520972   Val Loss:
2072.96792187209
Iterasi ke- 461   Train Loss:  2029.7032165122362   Val Loss:
2072.379158527575
Iterasi ke- 462   Train Loss:  2029.051964923123   Val Loss:
2071.79326768358
Iterasi ke- 463   Train Loss:  2028.4038131596506   Val Loss:
2071.2102355005404
Iterasi ke- 464   Train Loss:  2027.7587464672765   Val Loss:
2070.6300482051624
Iterasi ke- 465   Train Loss:  2027.1167501616885   Val Loss:
2070.0526920901066
Iterasi ke- 466   Train Loss:  2026.4778096284665   Val Loss:
2069.478153513675
Iterasi ke- 467   Train Loss:  2025.8419103227532   Val Loss:
2068.9064188994967
```

```
Iterasi ke- 468   Train Loss:  2025.2090377689217   Val Loss:
2068.3374747362136
Iterasi ke- 469   Train Loss:  2024.5791775602463   Val Loss:
2067.771307577176
Iterasi ke- 470   Train Loss:  2023.9523153585744   Val Loss:
2067.2079040401254
Iterasi ke- 471   Train Loss:  2023.3284368939997   Val Loss:
2066.6472508068955
Iterasi ke- 472   Train Loss:  2022.707527964538   Val Loss:
2066.0893346231
Iterasi ke- 473   Train Loss:  2022.0895744358038   Val Loss:
2065.5341422978313
Iterasi ke- 474   Train Loss:  2021.4745622406883   Val Loss:
2064.9816607033563
Iterasi ke- 475   Train Loss:  2020.8624773790393   Val Loss:
2064.4318767748146
Iterasi ke- 476   Train Loss:  2020.2533059173427   Val Loss:
2063.8847775099193
Iterasi ke- 477   Train Loss:  2019.6470339884045   Val Loss:
2063.340349968657
Iterasi ke- 478   Train Loss:  2019.0436477910364   Val Loss:
2062.798581272992
Iterasi ke- 479   Train Loss:  2018.4431335897402   Val Loss:
2062.259458606568
Iterasi ke- 480   Train Loss:  2017.8454777143972   Val Loss:
2061.7229692144165
Iterasi ke- 481   Train Loss:  2017.2506665599544   Val Loss:
2061.189100402661
Iterasi ke- 482   Train Loss:  2016.6586865861168   Val Loss:
2060.657839538229
Iterasi ke- 483   Train Loss:  2016.0695243170378   Val Loss:
2060.1291740485553
Iterasi ke- 484   Train Loss:  2015.4831663410139   Val Loss:
2059.6030914213
Iterasi ke- 485   Train Loss:  2014.8995993101776   Val Loss:
2059.0795792040585
Iterasi ke- 486   Train Loss:  2014.318809940196   Val Loss:
2058.5586250040733
Iterasi ke- 487   Train Loss:  2013.7407850099655   Val Loss:
2058.0402164879492
Iterasi ke- 488   Train Loss:  2013.1655113613135   Val Loss:
2057.524341381375
Iterasi ke- 489   Train Loss:  2012.5929758986965   Val Loss:
2057.010987468834
Iterasi ke- 490   Train Loss:  2012.023165588904   Val Loss:
2056.500142593331
Iterasi ke- 491   Train Loss:  2011.4560674607599   Val Loss:
2055.9917946561045
Iterasi ke- 492   Train Loss:  2010.8916686048294   Val Loss:
```

```
2055.485931616356
Iterasi ke- 493   Train Loss:  2010.3299561731235   Val Loss:
2054.98254149097
Iterasi ke- 494   Train Loss:  2009.7709173788073   Val Loss:
2054.4816123542364
Iterasi ke- 495   Train Loss:  2009.2145394959084   Val Loss:
2053.983132337582
Iterasi ke- 496   Train Loss:  2008.6608098590273   Val Loss:
2053.4870896292905
Iterasi ke- 497   Train Loss:  2008.1097158630494   Val Loss:
2052.9934724742393
Iterasi ke- 498   Train Loss:  2007.561244962858   Val Loss:
2052.5022691736212
Iterasi ke- 499   Train Loss:  2007.0153846730484   Val Loss:
2052.0134680846804
Error Trainig:  4012.9442451352884
Error Testing:  3883.722088811372
```

####PERTANYAAN 5

1. Berapa Error Memorization dan Generalization yang didapat?
2. Bandingkan dengan ketika kita gunakan seluruh fiturnya, mana yang lebih baik?
3. Lakukan eksperimen dengan menggunakan kombinasi fitur lainnya, seperti pada tabel berikut, dan catat hasilnya. Gunakan learning rate dan iterasi yang sama.
4. Mana yang terbaik? apa kesimpulan Anda?
5. Mengapa multiple feature dapat membantu?

JAWABAN 5:

1. Train MSE: 3785.76, Test MSE: 3788.91
2. Menggunakan seluruh fitur jauh lebih baik (Train MSE: 313.51, Test MSE: 319.17).
3. Tabel Hasil Eksperimen (dengan LR=0.5, Iters=1000)

Eksperimen	Fitur	MSE Train	MSE Test
1	seluruhnya	313.51	319.17
2	1	3785.76	3788.91
3	2	7264.93	7773.02
4	3	9783.20	9656.87
6	1,2	871.24	976.89
7	1,3	3416.93	3092.10
8	2,3	6776.64	7543.52
9	1,2,3	478.60	501.76

1. Model dengan menggunakan seluruh fitur memberikan hasil terbaik. Kombinasi banyak fitur memberikan informasi harga yang lebih komprehensif.
2. Multiple feature lebih baik karena merepresentasikan karakteristik target value (harga) dari sudut pandang yang berbeda, memperjelas batasan keputusan regresi dan mengidentifikasi pola kompleks yang tidak tertangkap oleh 1 fitur.

##4. Eksperimen Overfitting dan Underfitting

Data yang kita gunakan adalah data sintesis. Fitur 1-4 merupakan fitur sungguhan, sedangkan fitur sisanya merupakan feature mapping dari fitur fitur sungguhan, bisa berupa kombinasi, atau bahkan pangkat tinggi.

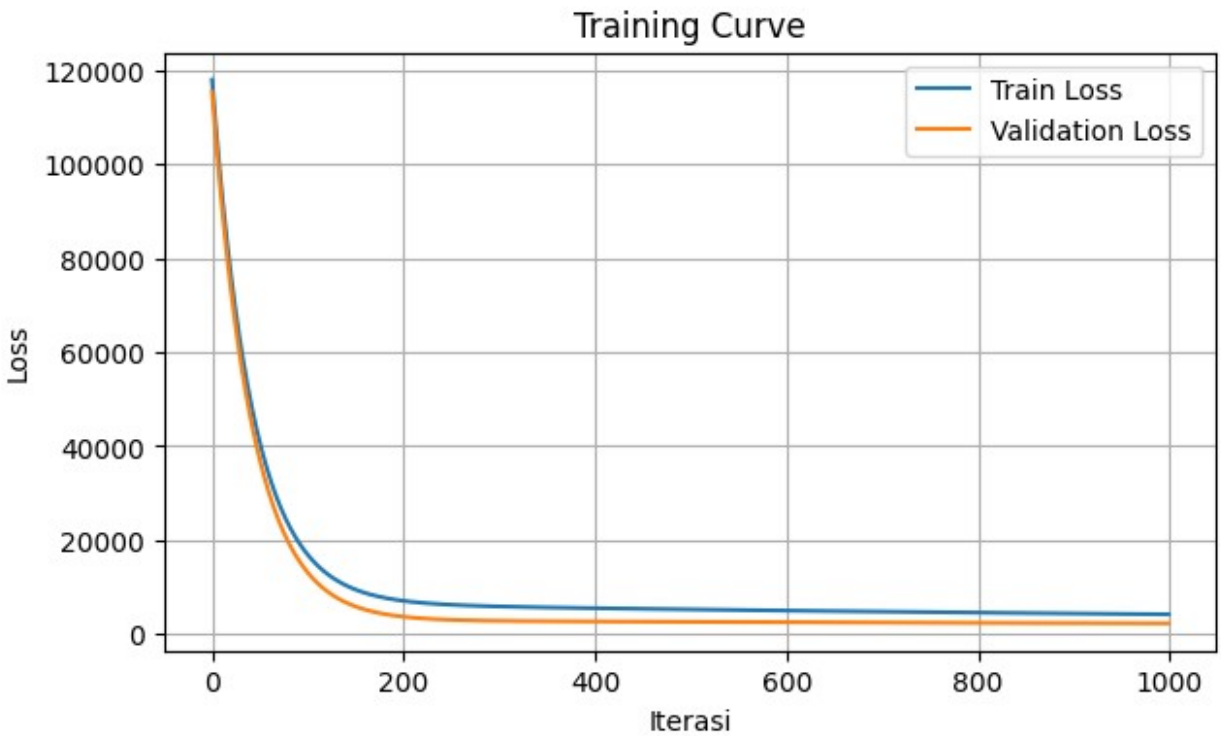
###a. Ambil data training yang tidak terlalu *banyak*

```
X_train_small, X_val_small, y_train_small, y_val_small =
train_test_split(X_train,y_train,0.99)
print("Shape data training: ", X_train_small.shape)
print("Shape data validation: ", X_val_small.shape)
```

```
Shape data training:  (12, 13)
Shape data validation:  (1188, 13)
```

###b. Training model dengan small data dan gunakan fitur-1 saja

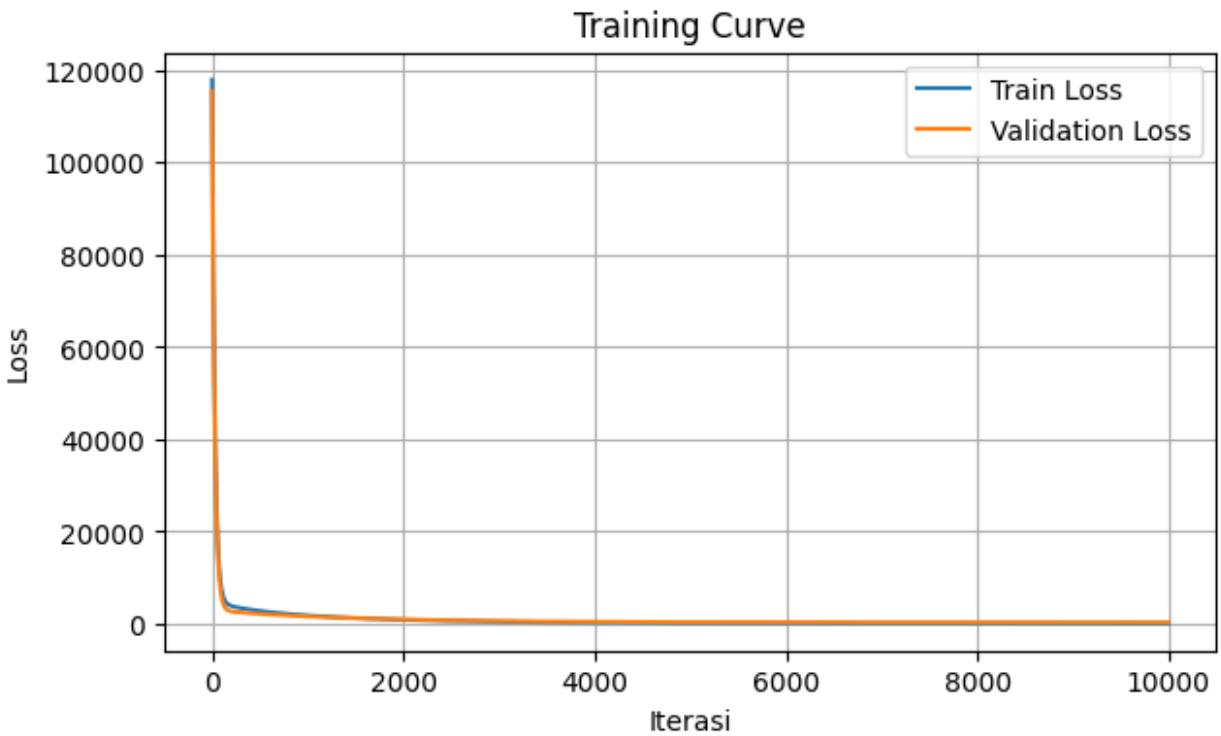
```
#Ambil fitur 1 saja
idx_fitur = [0,1]
#Train Model
w, hist_train, hist_val = train(X_train_small[:,idx_fitur],
y_train_small, X_val_small[:,idx_fitur], y_val_small, 0.01, 1000,
False)

#Plot History Train dan Validation Loss
plot_history_train_validation(hist_train, hist_val)

#Evaluasi dengan data training
hasil_prediksi = predict(w,X_train_small[:,idx_fitur] )
error = mse(y_train_small, hasil_prediksi)
print("Error Trainig: ", error)

#Evaluasi dengan data TEsting
hasil_prediksi = predict(w,X_test[:,idx_fitur] )
error = mse(y_test, hasil_prediksi)
print("Error Testing: ", error)
```




```
Error Trainig:  8427.370868920381
Error Testing:  4398.0271411596195
```

###c. Training model dengan small data dan gunakan fitur 1-4

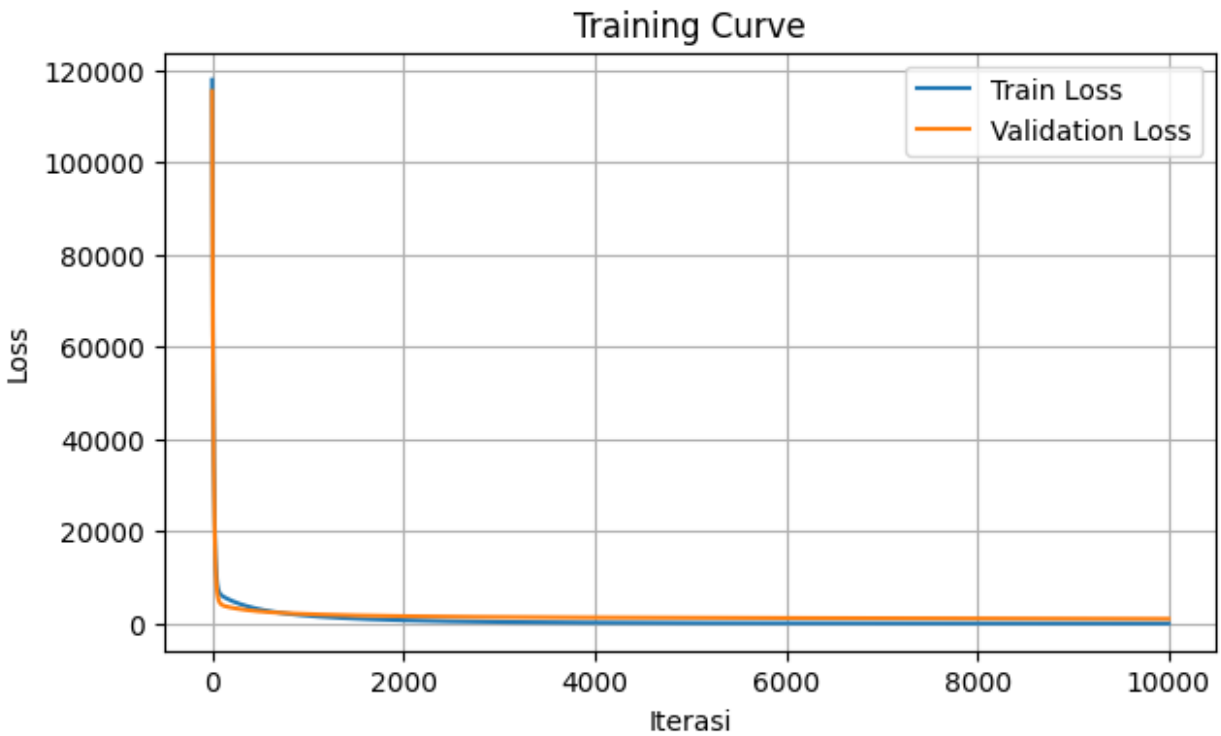
```
#Fitur yang digunakan
idx_fitur = [0,1,2,3,4]
#Train Model
w, hist_train, hist_val = train(X_train_small[:,idx_fitur],
y_train_small, X_val_small[:,idx_fitur], y_val_small, 0.01, 10000,
False)

#Plot History Train dan Validation Loss
plot_history_train_validation(hist_train, hist_val)

#Evaluasi dengan data training
hasil_prediksi = predict(w,X_train_small[:,idx_fitur] )
error = mse(y_train_small, hasil_prediksi)
print("Error Trainig: ", error)

#Evaluasi dengan data TEsting
hasil_prediksi = predict(w,X_test[:,idx_fitur] )
error = mse(y_test, hasil_prediksi)
print("Error Testing: ", error)
```




```
Error Trainig:  257.82024111773904
Error Testing:  735.7506721475415
```

###d. Training model dengan small data dan gunakan seluruh fitur

```
#Train Model
w, hist_train, hist_val = train(X_train_small, y_train_small,
X_val_small, y_val_small, 0.01, 10000, False)

#Plot History Train dan Validation Loss
plot_history_train_validation(hist_train, hist_val)

#Evaluasi dengan data training
hasil_prediksi = predict(w,X_train_small )
error = mse(y_train_small, hasil_prediksi)
print("Error Trainig: ", error)

#Evaluasi dengan data TEsting
hasil_prediksi = predict(w,X_test )
error = mse(y_test, hasil_prediksi)
print("Error Testing: ", error)
```




```
Error Trainig:  156.91291872995868
Error Testing:  2108.67874864009
```

####PERTANYAAN 6

1. Berapa Error Memorization dan Generalization yang didapat pada poin b, c, dan d?
2. Masalah apa yang terjadi pada poin b, d, kenapa hasilnya tidak sebaik pada poin c?
3. Berdasarkan pengamatan Anda pada plot history training dan validation apa ciri-ciri gejala underfitting? bagaimana hubungannya dengan jml iterasi, jml data, dan jumlah fitur yang digunakan?
4. Berdasarkan pengamatan Anda pada plot history training dan validation apa ciri-ciri gejala overfitting? bagaimana hubungannya dengan jml iterasi, jml data, dan jumlah fitur yang digunakan?

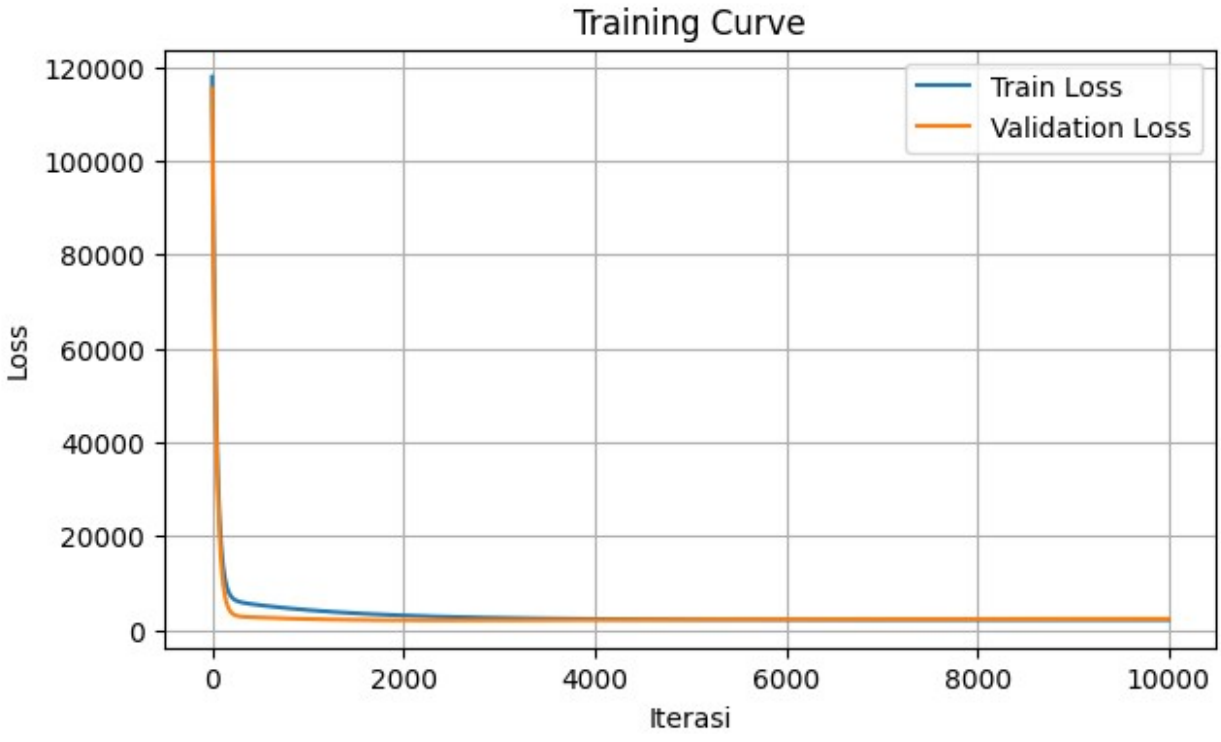
JAWABAN 6:

1. Error percobaan small data:
   - Poin b (1 fitur): Train = 8427.37, Test = 4398.03
   - Poin c (5 fitur): Train = 257.82, Test = 735.75
   - Poin d (Semua fitur): Train = 156.91, Test = 2108.68
2. Poin b mengalami underfitting (error sangat tinggi meskipun di test data). Poin d mengalami overfitting (training kecil tapi hasil test besar). Poin c lebih ideal karena jumlah fitur cukup pas dengan jumlah data.
3. Gejala underfitting: Loss training dan validation stabil di angka yang sangat tinggi. Berhubungan erat dengan model yang kurang kompleks (fitur terlalu sedikit), iterasi seringkali gagal konversikan error.

4. Gejala overfitting: Loss training lambat laun terus rendah dan sangat kecil (hapal data), tapi validation errornya mandek dan malah bertambah besar lagi, jarak antar error ini semakin membesar (gap margin). Berhubungan saat model memiliki rasio fitur yang terlampau kompleks melampaui variasi dan jumlah baris data observasinya (small data).

##5. Eksperimen Feature Selection untuk Memperbaiki Overfitting

```
#Fitur yang digunakan
idx_fitur = [0,1]
#Train Model
w, hist_train, hist_val = train(X_train_small[:,idx_fitur],
y_train_small, X_val_small[:,idx_fitur], y_val_small, 0.01, 10000,
False)

#Plot History Train dan Validation Loss
plot_history_train_validation(hist_train, hist_val)

#Evaluasi dengan data training
hasil_prediksi = predict(w,X_train_small[:,idx_fitur] )
error = mse(y_train_small, hasil_prediksi)
print("Error Trainig: ", error)

#Evaluasi dengan data TEsting
hasil_prediksi = predict(w,X_test[:,idx_fitur] )
error = mse(y_test, hasil_prediksi)
print("Error Testing: ", error)
```



```
Error Trainig:  4176.035948046695
Error Testing:  4858.017495590012
```

####PERTANYAAN 7 **Catatan**: Hanya ubah index fiturnya saja, jangan ubah parameter lain

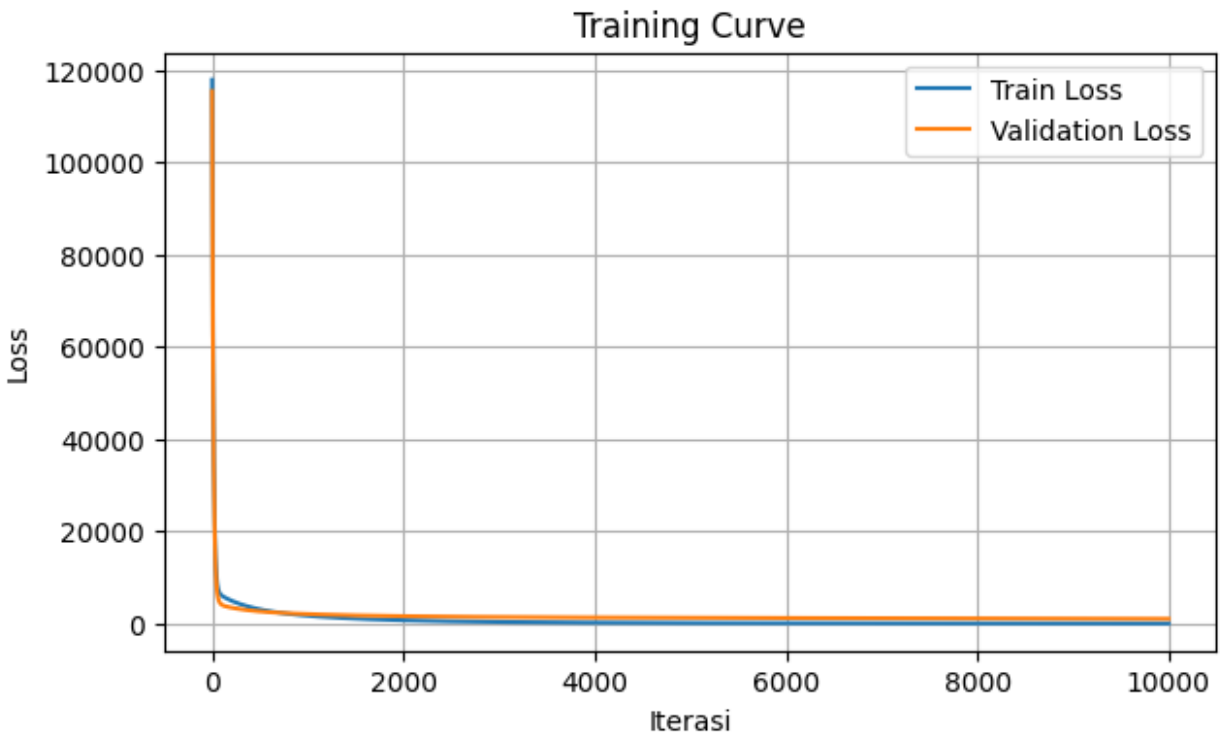
1. Lakukan sequential forward feature selection (SFS), untuk memilih fitur mana yang terbaik untuk digunakan, dengan cara melakukan training mulai dari 1 fitur, lalu menambahkan fitur yang paling meningkatkan performa model pada setiap langkah hingga semua fitur dicoba. Berdasarkan SFS kombinasi fitur mana yang terbaik?
2. Lakukan sequential backward feature selection (SBFS), untuk memilih fitur mana yang terbaik untuk digunakan, dengan cara melakukan training mulai dari seluruh fitur, lalu menghapus fitur yang paling menurunkan performa model pada setiap langkah hingga semua fitur dicoba. Berdasarkan SBFS kombinasi fitur mana yang terbaik?

JAWABAN 7:

1. Kombinasi fitur SFS terbaik: [1, 2, 3, 4, 5, 7, 12, 8] dimana iterasi setelahnya malah membuat model menjadi buruk (tidak mengurangi validation MSE terbaik 552.06).
2. Kombinasi fitur SBFS terbaik: Menghapus 8, 6, 11, 9, 10 menghasilkan validation MSE terbaik 566.12. Sisa fitur yang patut dipertahankan adalah: [1, 2, 3, 4, 5, 7, 12].

##6. Eksperimen Regularization untuk Memperbaiki Overfitting

```
#Gunakna semua fitur
idx_fitur = [0,1,2,3,4,5,6,7,8,9,10,11,12]
#Train Model
lamda = 0.0 #parameter regularizatio
w, hist_train, hist_val = train(X_train_small[:,idx_fitur],
y_train_small, X_val_small[:,idx_fitur], y_val_small, 0.01, 10000,
False,lamda)

#Plot History Train dan Validation Loss
plot_history_train_validation(hist_train, hist_val)

#Evaluasi dengan data training
hasil_prediksi = predict(w,X_train_small[:,idx_fitur] )
error = mse(y_train_small, hasil_prediksi)
print("Error Trainig: ", error)

#Evaluasi dengan data TEsting
hasil_prediksi = predict(w,X_test[:,idx_fitur] )
error = mse(y_test, hasil_prediksi)
print("Error Testing: ", error)
```




```
Error Trainig:  156.9129187299596
Error Testing:  2108.6787486400904
```

####PERTANYAAN 8 **Catatan**: Hanya ubah index fiturnya saja, jangan ubah parameter lain

1. Coba gunakan lamda= 0.0001, 0.001, 0.01, 0.1, 1, 10, 100. Mana yang memberikan hasil terbaik?
2. Apa yang dapat Anda simpulkan?

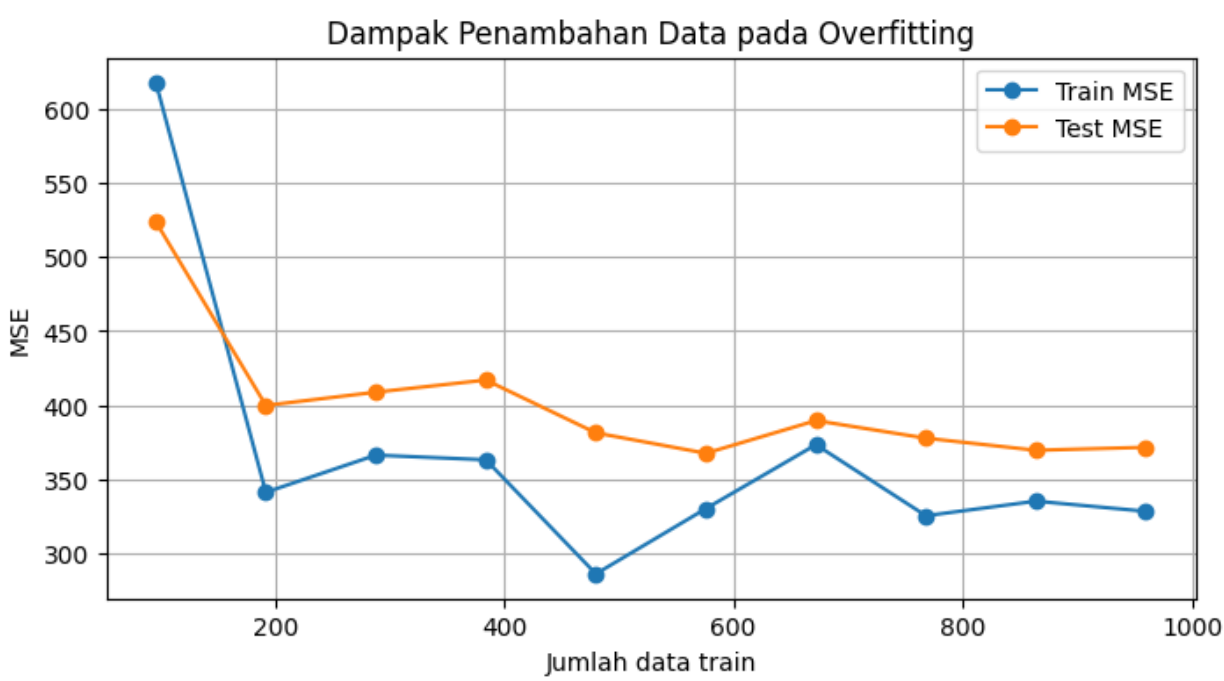
JAWABAN 8:

1. Lamda = 0.01 memberikan hasil test generalisasi yang terbaik (Train MSE: 170.55, Test MSE: 2097.85) dibanding tidak menggunakan lamda sama sekali (Train: 156.91, Test: 2108.68).
2. Regularisasi dapat mengurangi efek overfitting dengan menekan/memberikan penalti pada bobot parameter regresi berlebih tanpa membuangnya seutuhnya. Namun jika lamda terlalu besar, model menjauh dari pola data dan error akan semakin besar karena model berubah menjadi terlalu kaku (underfit).

##7. Eksperimen Melihat Efek perubahan jumlah data

```
rasio_jml_data_training = [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8,
0.9, 1]
results = []
#Train dan Test Model dengan jml data training sesuai rasio
for rasio in rasio_jml_data_training:
```

```
  jml_data = int(rasio*X_train_sungguhan.shape[0])
  X_train_rasio, y_train_rasio =
pilih_sample_random(X_train_sungguhan, y_train_sungguhan, jml_data)
  w, hist_train, hist_val = train(X_train_rasio, y_train_rasio, X_val,
y_val, 0.01, 10000, False)

  #Evaluasi dengan data training
  hasil_prediksi = predict(w,X_train_small )
  error_train = mse(y_train_small, hasil_prediksi)

  #Evaluasi dengan data TEsting
  hasil_prediksi = predict(w,X_test )
  error_test = mse(y_test, hasil_prediksi)

  results.append((jml_data, error_train, error_test))

jmls = [r[0] for r in results]
err_train = [r[1] for r in results]
err_test = [r[2] for r in results]

plt.figure(figsize=(8,4))
plt.plot(jmls, err_train, marker="o", label="Train MSE")
plt.plot(jmls, err_test, marker="o", label="Test MSE")
plt.xlabel("Jumlah data train")
plt.ylabel("MSE")
plt.title("Dampak Penambahan Data pada Overfitting")
plt.legend()
plt.show()
```



####PERTANYAAN 9

1. Berdasarkan grafik, apa kesimpulan Anda terhadap efek penambahan jumlah data terhadap kapasitas model?

JAWABAN 9:

1. Semakin banyak jumlah data yang tersedia atau dilatihkan, gap antara training error dan test error (generalization gap) semakin rapat/mengecil dan membaik. Penambahan jumlah data dapat membantu mencegah efek overfitting karena seiring besarnya kapasitas data model dapat berlatih keragaman data dengan jauh lebih baik tanpa mudah terjebak merapal memorisasi data sedikit.